Contrasting growth responses to longer warmer growing seasons between juvenile and mature trees

by

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The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

Contrasting growth responses to longer warmer growing seasons between juvenile and mature trees

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Abstract

Anthropogenic climate change affects natural systems at the global scale. The most frequently observed biological impact of climate change—shifts in the timing of recurring life history events (phenology)—is likely to have additional effects. For trees, shifted phenology has extended the vegetative growing season, which is widely expected to increase tree growth, with important effects on forest carbon sequestration dynamics. However, multiple recent studies have failed to find this relationship and suggested that shifts in drought or competition may prevent increased growth. Here, we leverage two unique datasets of annual vegetative phenology and growth (via tree rings) data, one from a common garden of four species of juvenile trees and the other from a community science program with eleven species of mature trees. Both of these projects took place in an urban arboretum where drought and competition are limited. We found an overall positive response to longer and warmer seasons for our juvenile trees, with little effect of provenance. For these, most species showed a stronger response to the end of season than the start of season, suggesting that we should look beyond the start of season when studying growth responses to longer growing season. However, our second study on mature trees supported recent results on the decoupling between growth and growing season length with little effect of life-history strategies. These findings further suggest that we should consider more than one year of growing conditions, as growth responses shifted for two species when considering the thermal season of the previous year. This may reveal hidden growth trends and, in addition, our results may highlight the need to better understand carbon allocation to storage and growth. Overall, our findings suggest that longer seasons with climate change can benefit forest regeneration and range expansion, but negatively impact the overall carbon storage of temperate forests.

Lay Summary

Climate change extends the time during which plants can grow through earlier springs and later autumns. For trees, a warmer and longer growing season means they should grow more. However, recent research showed that this may not be the case. Here we ask if juvenile (Chapter 1) and mature (Chapter 2) broadleaf trees grow more with longer and warmer growing seasons. Our results show that while young trees grew more and benefited from warmer and longer seasons, adult trees did not. This has important implications because young trees are crucial for forest regeneration, but they hold a relatively insignificant proportion of the carbon stored in forests compared to adult trees. Our results thus partially support the recently observed decoupling between season length and growth, which could have important implications for future carbon sequestration by forests.

Preface

For my first chapter, I leveraged phenology data that was collected by Dr. D.M. Buonioto and Dr. C.J. Chamberlain in Boston, which was part of a project started by Dr. E.M. Wolkovich. The research project idea was proposed by E.M. Wolkovich and conducted by me with her help throughout. I processed, scanned, and measured the tree cross-sections and cores that were collected by Dr. Neil Pederson, D.M. Buonaiuto, Dr. Deirdre Loughnan and E.M. Wolkovich. I performed the analyses and created the figures with the help of Dr. Victor Van der Meersch and E.M. Wolkovich. Charlotte Capistran Champoux helped with improving the figures. I did not use generative artificial intelligence (GenAI) to write this thesis for any creative or idea generation process. I used Claude for performing simple repetitive coding tasks, including help with the figure coding, data cleaning and improving coding workflow efficiency (Anthropic, 2026). This work will be submitted for publication soon. E.M. Wolkovich and I wrote the manuscript, and the following co-authors reviewed and edited it: Victor Van der Meersch, D.M. Buonaiuto, C.J. Chamberlain, Deirdre Loughnan, Neil Pederson.

My second chapter partially consists of leveraging phenology data that was collected by the Tree Spotters, which is a citizen science program that E.M. Wolkovich started with C.J. Chamberlain in 2015. I also presented the results of Chapter 2 to the Tree Spotters citizen scientists in March 2026. The project idea was broadly proposed by E.M. Wolkovich and conducted by me; I also helped expand the project questions to consider wood anatomy and life history strategies. I collected the tree cores with Xiaomao Wang's help, and I processed and measured them. I also performed the analyses with Dr. Victor Van der Meersch and E.M. Wolkovich. I used GenAI the same way I did for Chapter 1. E.M. Wolkovich and I wrote the manuscript, and this work will be submitted for publication.

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Acknowledgments

Going through a career change as drastic as being a cook, then a server, to pursuing graduate studies in ecology is somewhat unusual, and the experience I brought from the restaurant industry to academia may not be obvious. However, I think the rigour and work ethic that my teachers and chefs have taught me during those eight years have forged me into a more curious, hard-working, and perseverant student.

Many people contributed to this thesis and made this degree more enriching. Lizzie in particular. Thank you for your support, guidance and trust through the last three years. From my beginning in the lab as an undergrad to the end of my MSc, you pushed me to always think further and question assumptions. Your mentoring in stats, writing, and scientific integrity laid a solid foundation that I will, without a doubt, carry through my career.

Thank you to all the Tree Spotters for your dedicated years of passionate phenology collection. You made this amazing dataset available to me that has been such a precious asset for my work. I acknowledge the insightful comments from my committee members, Sally Aitken and Sean Michaletz, that helped improve this thesis and Rob Guy for your perspectives on plant physiology. Thank you also to all the undergrads who helped me with the data collection of my experiment, and especially to Ken for your support in stats and coding.

This two-year journey would not have been as pleasant and enriching without my wonderful and brilliant lab colleagues. Thank you to Mao for always listening to me, whether it was for a 2-minute water run or for many hours during fieldwork. To Victor, for your teachings and great ideas, and the occasional roast of my French! Deirdre, for all your advice, connections, and academic street knowledge. Fredi, for your trust and teachings early on during my tree growth journey. Britany, Justin and Avery, thank you for the wonderful memories from our time at Mount Rainier and beyond! Finally, thank you to my family, my Vancouver and Montreal friends, and especially to Charlotte. Your constant and warm support through my ups and downs is more important to me than you can imagine.

Chapter 1

Warmer and longer growing seasons increase growth for juvenile trees in a common garden

1.1 Introduction

The most frequently observed biological impact of climate change is shifts in phenology—the timing of recurring life history events (Calvin *et al.*, 2023; Menzel *et al.*, 2006; Parmesan *et al.*, 1999). Across many plant species, spring events have advanced (Chmielewski & Rötzer, 2001; Fu *et al.*, 2014; Wolfe *et al.*, 2005), while autumn events have delayed—but to a much lesser extent (Gallinat *et al.*, 2015; Jeong & Medvigy, 2014). Together, early springs and later autumns lengthen the growing season, which should increase growth for many species, including trees (Keenan *et al.*, 2014; Stridbeck *et al.*, 2022).

How much trees increase growth with longer seasons is important given the major role of forests in carbon sequestration (Friedlingstein *et al.*, 2026). Temperate forests account for roughly 40% of the annual global carbon sink, diminishing the consequences of human-emitted greenhouse gas emissions on the climate (Gibbs *et al.*, 2025). With warming, research has repeatedly found a connection between longer seasons and increased carbon storage (Boisvenue & Running, 2006; Finzi *et al.*, 2020; Gao *et al.*, 2022). The future of this net carbon sink by temperate forests, however, relies on the assumption that longer and warmer seasons will keep increasing carbon storage by trees (Pan *et al.*, 2024). A shift in this relationship could impact the future of this critical carbon sink.

New studies suggest that longer growing seasons do not always increase growth (Cabon *et al.*, 2022; Charlet De Sauvage *et al.*, 2022; Dow *et al.*, 2022; Matula *et al.*, 2023; Silvestro *et al.*, 2023). Recently, Dow *et al.* (2022) found that warmer springs lengthened the growing season, but did not increase the peak growth rate or total annual growth in two temperate forest communities. Working across biomes, Cabon *et al.* (2022) found a similar trend, where gross primary production and growth appeared decoupled. However, neither of these studies could clearly establish why they found no relationship, while previous studies have. Understanding whether temperate forests will persist as a net carbon sink requires answering why trees do not grow more despite longer growing seasons.

A longer growing season does not necessarily mean more favorable thermal conditions for growth. While the well-documented advance of spring phenology elongates the growing season (Keenan *et al.*, 2014), spring days are short and usually cool (Figure 1.1a). Therefore, this advance may contribute little to a tree's annual growing degree days (GDD) accumulation (Figure 1.1b). In contrast, even small shifts to later end of season events can lead to larger changes in annual GDD; since more GDD accumulates per day during this period, any extra end of season days may be more important than additional start of season days (Buonaiuto *et al.*, 2026).

Other factors beyond the thermal conditions may affect whether longer seasons increase growth. While trees may have a longer period to grow, water availability may become insufficient to support continuous growth throughout the growing season (Wang *et al.*, 2025). Competition for light and nutrients may further limit the capacity of trees to benefit from longer seasons, especially for the ones positioned below the forest canopy (Augspurger & Bartlett, 2003; Cleland & Wolkovich, 2024; Sweeney *et al.*, 2024). Additionally, trees may have internal constraints that prevent them from increasing their growth with longer seasons. Growth, like other biological processes, is under strict genetic control. Thus, even with optimal environmental conditions, physiological constraints may prevent temperate trees from growing indefinitely (Körner *et al.*, 2023; Wolkovich *et al.*, 2025), but we currently have a limited understanding of how well tree phenology and growth can track changing conditions.

Current research suggests that leaf and wood phenology differ in their plasticity in response to changing environmental conditions across species and life stages (Dox *et al.*, 2022). Leaf phenology (e.g. leafout) in the spring is a highly plastic trait with substantial variation across years with different weather, and also across species. Within a single community leafout can vary across species by over a month, with a general trend of evergreen trees and species with more conservative growth strategies leafing out later (Loughnan *et al.*, 2025). Additionally, leaf phenology and growth appear closely synchronized in diffuse-porous species (e.g., *Betula;* Čufar *et al.*, 2008) and less so in ring-porous species (e.g., *Oaks*; González-González *et al.*, 2013), making leaf phenology a good proxy for setting the start and end of the growing season in angiosperm diffuse-porous species (Marchand *et al.*, 2021; Silvestro *et al.*, 2025; Suzuki *et al.*, 1996). While this proxy is useful for setting the boundaries of the growing season, it does not indicate how much trees grow annually. How flexible growth is in response to changing growing season length and conditions varies between species and populations, and may also differ with tree age, with juvenile trees being generally more flexible and responsive than their adult forms (Augspurger & Bartlett, 2003; Way & Montgomery, 2015).

Local adaptation could limit tree growth responses of different populations to future climate (Soolanayakanahally *et al.*, 2013; Wolkovich *et al.*, 2025). While leaf phenology (e.g. leafout) in the spring is highly plastic and driven by temperature, end of season phenology (e.g. budset) is much less plastic and mostly driven by photoperiod (Baumgarten *et al.*, 2026; Singh *et al.*, 2017), with evidence of local adaptation (Zeng & Wolkovich, 2024). Provenance trials have repeatedly found trees from northern latitudes set bud earlier, even when grown in warm climates with longer growing seasons (Aitken & Bemmels, 2016; Way & Montgomery, 2015). This local adaptation and related reliance on photoperiod to trigger budset (an end of season event) likely explain why shifts in EOS with warming have been so much smaller than shifts in start of season. This predicts a latitudinal gradient in growth responses, with more limited responses in poleward populations where the end of season is constrained to happen earlier in the season. Thus, common gardens investigating the effect of provenance latitude could help predict how climate change will affect tree growth.

To understand the relationship between growing seasons and growth, we need to understand how growth varies across species and populations while disentangling how much longer seasons are also more thermally favorable seasons. Here, we address this by studying how growth and growing seasons relate across species, populations and multiple metrics of growing season length. Working with four juvenile tree species sourced from four populations and grown in one common garden, we examine how growth, measured via tree rings, and leaf phenology covary over three years. We consider effects of both the length of the season (i.e. number of growing days in the season) and its thermal conditions (i.e. GDD accumulated in the season).

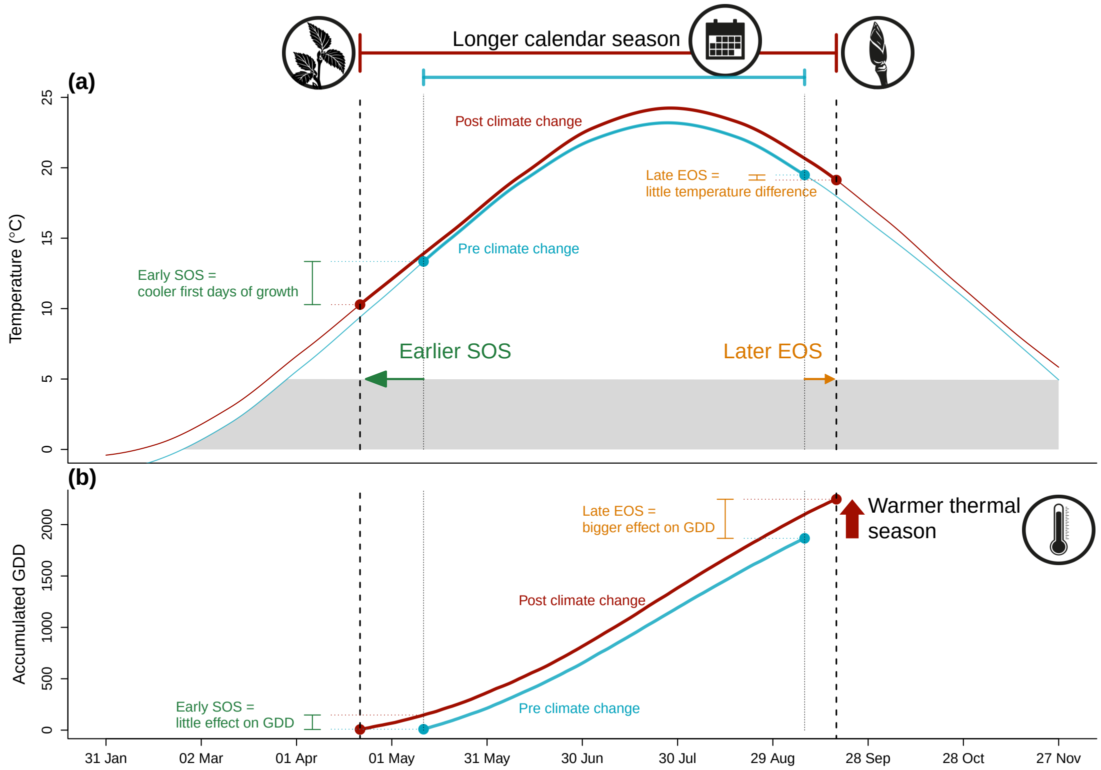

Figure 1.1: Shifting growing season lengths with climate change affects both the temperature trees experience throughout the growing season (a) and how many growing degree days (GDD) they accumulate (b). Here we show changes in daily (smoothed) and accumulated temperatures before (1945-1965, in blue) and significant anthropogenic climate warming (2005-2025, in red) using data local to our study. The arrows represent hypothetical shifts in spring (green) and fall (orange) phenology. With climate change, spring events have advanced and fall phenology has delayed, but to a much lesser extent—lengthening the calendar growing season. This concurrently changed the temperature trees experience early in their season, with cooler first days of growth (a) causing little accumulation of GDD (b). In contrast, weaker shifts in end of season and little temperature change (a) result in a relatively high GDD accumulation (b) due to warmer temperatures at this time of the season. Alongside these start and end of season shifts, warmer summer temperatures increased GDD accumulation, further warming their thermal seasons. We used mean daily temperature data from the Boston Logan International Airport Climate station for the smoothed temperature and GDD curves (, NOAA).

1.2 Materials and Methods

1.2.1 Common garden setup and data collection

We collected seeds from 18 species of deciduous trees and shrubs from four provenances across a 3.5° latitudinal gradient (Figure 2.1; Table A.1, A.2) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings (Baskin & Baskin, 2014). In the spring of 2017, we planted the seedlings outside at the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned for airflow approximately 1.5 meters above the ground in the fall of 2020 (Buonaiuto *et al.*, 2026).

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December; the exact frequency varied within and across years (for more details, see Figure A.1). For this, we used a modified BBCH scale to observe two phenological stages: leafout, which we defined as: BBCH 15 and budset as BBCH 102 (proposed phenophases in Buonaiuto *et al.*, 2026; Finn *et al.*, 2007). We had a total of 239 leafout and 230 budset observations (which we refer to as the 'full datasets'), but for some individual trees in certain years we did not have both events; thus our total of matched full growing season duration (which we refer to as the 'restricted dataset') is 227 observations across 81 individuals.

In the spring of 2023, we collected tree ring samples from tree species with high survivorship. Thus, we have four species in the *Betulaceae* family: grey alder (*Alnus incana*), yellow birch (*Betula alleghaniensis*), paper birch (*Betula papyrifera*) and grey birch (*Betula populifolia*). We collected cross-sections on 78 individuals by cutting the trunk 2 cm above the root collar, and cores on 43 individuals using standard dendrochronological methods (Cook & Kairiukstis, 1990), for a total of 81 individual trees (Table A.1). We stored the cores in modified (with holes) plastic straws to promote aeration and left them to dry with the cross-sections at ambient temperature for three months.

1.2.2 Sample processing, imaging and measuring

We mounted cores on wooden mounts, and sanded both the cores and cross-sections using progressively finer sandpaper grits (150, 300, 400, 600, 800, 1000), then scanned the cores and cross-sections at a resolution of 6250 dpi with a high-resolution tree ring scanner (Fong *et al.*, 2026). Using the digitized images, we measured ring widths with ImageJ (Schneider *et al.*, 2012) at three standard locations for each cross-section (at 120° from each other, averaging the three measurements to yield one ring width per year, per individual), and one ring width for each core. Since trees do not grow symmetrically, having three measurements at different angles is better than one. Therefore, when we had cross-sections ($n = 78$) and cores ($n = 43$) for an individual, we used only the measurements from the cross sections ($n = 81$).

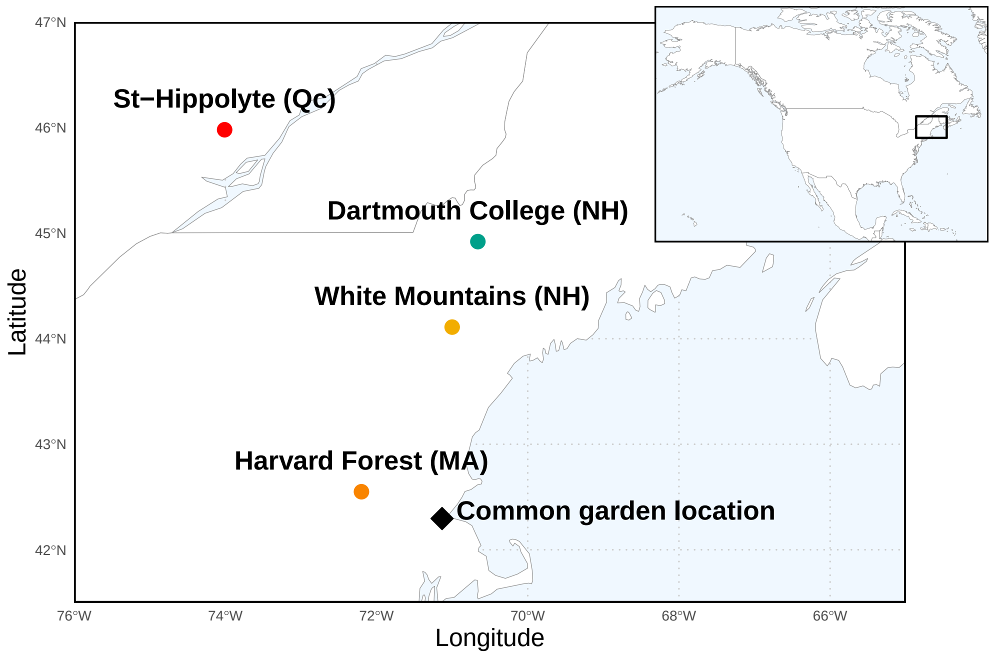

Figure 1.2: Provenances where the seeds were sourced from are depicted as circles, while the diamond denotes the location of the common garden. The common garden location is where the study took place at the Weld Research Building (Boston, MA). The colors of the dots match the model estimates of Figure A.3. We provide climate variables in Table A.1.

We performed visual cross-dating. This consists of assigning a calendar year to each ring and comparing the year-to-year growth variation and anomalies across replicates of the same species (Cook & Kairiukstis, 1990). We did not perform statistical crossdating or detrending because our chronologies (number of years of data) were extremely short (Raden *et al.*, 2020) and showed no evidence of consistent allometric trends (Figure A.2). We controlled for effects of different years by including 'year' in our models (see equation 2.1 in 'Data analyses' below).

1.2.3 Data analyses

To understand how growth shifted with season length, we used Bayesian hierarchical models, estimating how growth, measured as ring width, covaried with season length, measured in four different ways: GDD, GSL, SOS and EOS. We calculated the thermal season using growing degree days (GDD; mean daily temperature above 5°C) accumulated between leafout and budset, and the calendar season (or growing season length; GSL) as the number of days between leafout and budset. Finally, we used leafout as the start (SOS) and budset end of the season (EOS) as separate predictors that represent the boundaries of the growing season. For interpretability, we report ring width change (log(mm)) per GDDs accumulated in an average week (seven days) in May (day of year 121 to 150 from 2018 to 2020), for the thermal season, and ring width change per week of season length, leafout and budset for GSL, SOS and EOS, respectively. We report effects when the

90% uncertainty intervals (UI) do not overlap with zero and consider a trend if the 50% UI do not overlap with zero.

For daily climate data across all the years of phenological sampling (2018-2020), we combined two sources. For 2018 and 2019, we used data from the climate station located at Weld Hill research station (Arnold Arboretum, 2026). Because the data were not available for that station after August 2020, we used data from the weather station 'Boston (Weld Hill)' of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (see Figure A.12 for the climate overlap between the two stations in 2020; Carroll *et al.*, 2026). We used the North American Drought Monitor Maps to determine drought occurrence across years at our study location and used the proposed drought intensity index to qualify the presence of moderate, severe or extreme droughts (National Centers for Environmental Information, 2026).

For each predictor, we estimated its effect on observed ring width (i, denoting each observation) for each tree ($treeid$) in each year ($year$) as:

$$\begin{aligned}
ln(ringwidth)_i &\sim Normal(\mu, \sigma_y) \\
\mu = \alpha + \alpha_{\text{species}} &+ \alpha_{\text{prov}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\
\alpha_{\text{prov}} &\sim Normal(0, \sigma_{\alpha\,\text{prov}}) \\
\alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha\,\text{treeid}})
\end{aligned} \tag{1.1}$$

This model estimates a species-specific effect of each season length predictor (e.g. GDD) on growth ($\beta_{species}$), while also controlling for other variation due to species ($\alpha_{species}$), provenance (α_{prov}), individual tree (α_{treeid}) and year (α_{year}). We ran the models using both natural units, and also using scaled predictors (via z-scoring; Gelman & Hill, 2006) to allow for direct comparison of effect sizes across different metrics. We report estimates as means and 90% uncertainty intervals (UI).

We used weakly informative priors for all models (increasing the priors by 2$\times$ for the z-scored models did not qualitatively affect the results). We ran four chains with 2000 warm-up and 2000 sampling iterations each, for a total of 8000 sampling iterations and checked diagnostics of output: chains had no divergent transitions, $\hat{R}$ values below 1.01, and effective sample sizes (n_{eff}) greater than 10% of the model iterations. We also fit the models for SOS and EOS using both the restricted (n = 227) and full datasets for each predictor (SOS n = 239 and EOS n = 230. We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R.

Because ring widths decrease with age—particularly with young trees—we also ran the models using basal area increment as the response variable only for the 78 replicates with cross-sections. This allows for a more representative portrayal of annual growth increment by using the total ring area, instead of its width alone. To calculate basal area increment, we summed each ring width separately at the three distinct locations on the cross sections, yielding three radii from pit to bark.

We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring: $\pi \times$ (inner radius^2 − outer radius^2), and averaged across the three BAI for each ring.

1.3 Results

At the common garden location, the mean summer temperature was similar across years, with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C). There was no drought for those three years, except for in 2020, when a moderate summer drought (June-August) ramped up to a severe drought in September (National Centers for Environmental Information, 2026). The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018; Figure A.10a and b). The earliest leafout was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020; Figure A.10c). The earliest budset was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018; Figure A.10d). Finally, annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018; Figure A.9).

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in % —henceforth '%/day' or '%/GDD,' with days and GDD being adjusted to represent more meaningful units, as described above (see 'Data analyses' section in Methods for more details). Warmer thermal seasons increased growth for paper birch (*Betula papyrifera*; β = 16.05, UI: 6.37, 26.05 %/GDD), grey birch (*Betula populifolia*; β = 7.23, UI: 2.33, 12.25 %/GDD), and yellow birch (*Betula alleghaniensis*; β = 4.22, UI: 0.45, 8.15 %/GDD), grey alder (*Alnus incana*) trended in the same direction (β = 2.18, UI: -1.1, 5.57 %/GDD); Figure 1.3a and e; TableA.3). Unlike its response to warmer thermal season, *A. incana* grew more with longer calendar seasons (β = 5.27, UI: 0.14, 10.66 %/day). *B. papyrifera* also grew more with longer calendar seasons (β = 20.7, UI: 5.39, 37.14 %/day), while the other two *Betula* did not, though they trended in the same way (*B. alleghaniensis*: β = 3.39, UI: -2.69, 9.74 %/day, and *B. populifolia*: β = 5.87, UI: -1.91, 13.99 %/day; Figure 1.3b and f; Table A.3). On a scaled axis (through z-scoring, Gelman & Hill, 2006), the thermal and calendar seasons were similarly predictive (Figure A.22). Results with basal area increment (BAI) for the GDD model were similar to ring width (see Table A.6).

In contrast to our expectation, we found most species grew more with a later end of season, and only one species grew more with an earlier start of season (*A. incana* (one of the two species that a longer calendar season increased growth for: β = -7.07, UI: -12.73, -1.14 %/day; Figure 1.3c and g; Table A.3). *B. populifolia* trended in the opposite direction (β = 5.05, UI: -6.93, 17.84 %/day; Figure 1.3c and g; Table A.3). *B. alleghaniensis*, which did not grow more with longer calendar season, grew less with an earlier start of season (β = 10.65, UI: 0.48, 21.60 %/day; Figure 1.3c and g; Table A.3)—but more with a later end of season (β = 12.22, UI: 5, 19.68 %/day). Alongside *B. alleghaniensis*, two other species grew more with a later end of season (*B. populifolia*: β = 10.17, UI: 2.65, 18.03 %/day, and *B. papyrifera*: β = 23.58, UI: 9.28, 39.01 %/day; Figure 1.3d and h; Table A.3), with the latter being the only one of those three that also grew more with longer calendar season. The estimates were unchanged when using the full (SOS: $n = 239$ and EOS: $n = 230$) or restricted datasets ($n = 227$; Figure A.5a).

We did not find any clear effect of year on growth (2018: $\alpha = 0.19$, UI: -0.77, 1.15 RW; 2019: $\alpha = 0.18$, UI: -0.79, 1.14 RW; 2020: $\alpha = -0.43$, UI: -1.40, 0.53 RW; Table A.8), and there were no apparent differences in baseline growth between species (*A. incana*: $\alpha = 1.06$, UI: -3.19, 5.21 RW; *B. alleghaniensis*: $\alpha = 0.21$, UI: -3.97, 4.34 RW; *B. papyrifera*: $\alpha = -4.00$, UI: -8.89, 0.75 RW; *B. populifolia*: $\alpha = -0.66$, UI: -4.94, 3.63 RW; Figure A.6 and A.7; Table A.8). In addition, growth did not vary with provenance (Dartmouth College (NH): $\alpha = -0.03$, UI: -0.22, 0.14 RW; Harvard Forest (MA): $\alpha = -0.10$, UI: -0.33, 0.07 RW; St-Hippolyte (Qc): $\alpha = 0.04$, UI: -0.14, 0.24 RW; White Mountains (NH): $\alpha = 0.08$, UI: -0.09, 0.29 RW; Figure A.3; Table A.4). The estimates did not change when fitting the species provenances individually (see Figure A.4 for details on this separate intercept-only model). The limited variation between provenances ($\sigma = 0.17$, UI: 0.02, 0.48 RW) was similar to variation across individual trees ($\sigma = 0.17$, UI: 0.05, 0.27 RW; Table A.8).

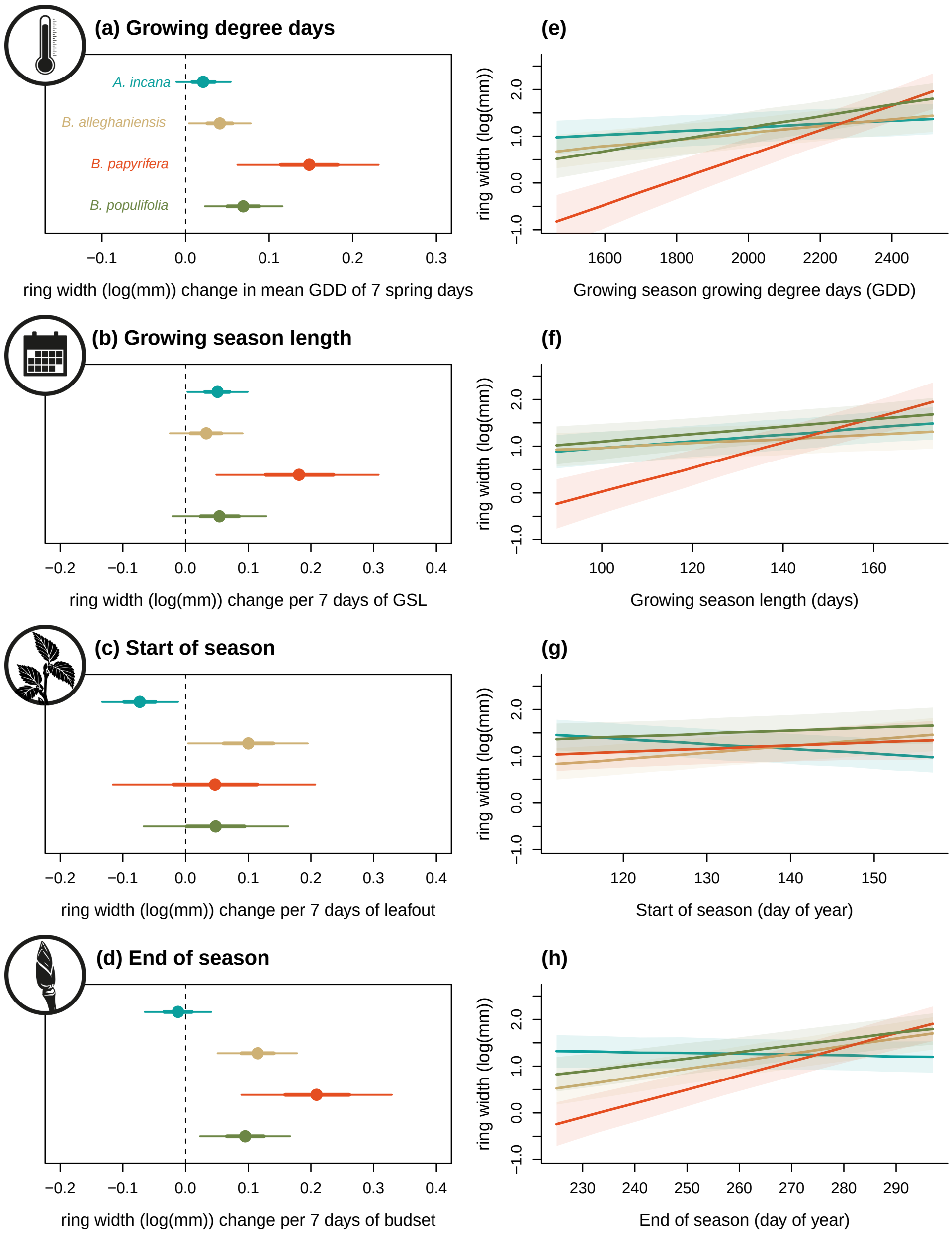

Figure 1.3: Estimates of ring width change (log(mm)) as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length, start of season, and end of season across four species: grey alder (*Alnus incana*), yellow birch (*Betula alleghaniensis*), paper birch (*Betula papyrifera*) and grey birch (*Betula populifolia*; species colors are the same across the panels). Panels on the left (a to d) represent the mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor, which we represent through ring width change in mm on the log scale per adjusted predictor (see 'Data analyses' section in Methods for more details on predictor adjustment). See Table A.3 for parameter estimates. Panels on the right (e to h) show the estimated response of ring width as a function of each predictor, with the lines depicting the mean response at each predictor value, and the shaded area, the 50% uncertainty intervals.

1.4 Discussion

Here we used 81 juvenile trees of four species from four provenances to study how longer growing seasons affect tree growth (using annual growth via tree rings and leaf phenology as a proxy for the start and end of season). Longer growing season increased growth for all species, but varied depending on the metric used to measure season length, with the thermal growing season more often relating to growth than the calendar season.

Our results support the idea that the thermal season is a better predictor for growth (Chuine & Régnière, 2017; Wang & Engel, 1998). By accounting for differences in daily temperatures, the thermal season may be more closely aligned with when and how fast plants can grow each day compared to the calendar season, which assumes a stable influence of each day on growth. Thus, the short and cool spring days early in the season count for less in the thermal season than the calendar season (Buonaiuto *et al.*, 2026). Further supporting this, when we disentangled the calendar season by using the start and end of season separately, we found a larger response of growth to the end of season, with little effect of the start of season.

The finding that the end of season appears more related to growth than the start of season is aligned with increasing work suggesting the start and end of season together may dynamically determine growth (Zohner *et al.*, 2023). This implies that research focusing solely on start of season (Dow *et al.*, 2022; Gao *et al.*, 2022; Wheeler *et al.*, 2016) could miss when longer seasons lead to more growth, and also that the strong focus on start of season in phenology research may ignore an important part of the season for growth. While studies have long called for more research on the end of season (Gallinat *et al.*, 2015; Gunderson *et al.*, 2012; Piao *et al.*, 2019; Richardson *et al.*, 2012), our results suggest we may especially need more work that monitors how growth and phenology shift across the season (Zani *et al.*, 2020; Zohner *et al.*, 2023). More studies using data with a finer temporal scale (e.g., daily or weekly data through dendrometers and/or xylogenesis) could confirm whether most growth occurs closer to the end of season versus the start of season. To date, these studies have shown strong species differences as to when most of the growth happens in the season. However, most of these studies were done on a small number of European and/or conifer species (e.g. Balducci *et al.*, 2019; Etzold *et al.*, 2022; Zweifel *et al.*, 2016). More data across species spanning a wider range of life history strategies could elucidate how dynamically growth can change throughout the season.

Our results suggest that different growth strategies between species may yield different responses to longer seasons. Our most ruderal species, *A. incana* was the only species to grow more with a longer calendar season and an earlier start of season(Grime, 1977). This suggests that species with more ruderal, flexible life-history strategies may actually take advantage of growing more in the spring despite cooler, shorter days (Bowman & Hacker, 2023; Furlow, 1997; Grime, 1977). In contrast, *B. papyrifera* and *B. populifolia*, which are fast growing, but more conservative than *A. incana*, were both most driven by the end of season, suggesting that they would be able to exploit these late season days characterized by warmer temperatures (Burns *et al.*, 1990). Lastly, our slowest growing, longest living species, *B. alleghaniensis*, responded to both the start and end of season. This points towards a possible 'neutralizing' response where an early start of season decreases growth, but a later end of season increases it, muting this species' response to a longer calendar season. Studies including an even wider span of growth strategies across species may find further differences. Our four study species all grow under closed canopy are relatively fast-growing, which should allow them to track changing conditions better than their slower-growing counterparts. While we found an overall positive trend of warmer seasons on growth, this might have changed if we used slower growing, more conservative species (such as *Quercus*; Burns *et al.*, 1990). Additionally, we used juvenile trees, which tend to be more reactive and could in part explain why they grew more with warmer seasons (Augspurger & Bartlett, 2003; Sweeney *et al.*, 2024). While other studies on mature trees have found the same result, our results may have been more muted if the trees were grown in a forest setting. We grew trees in a common garden setting with potentially more space between trees than in a natural setting (spacing was between 1.5 and 3 m), without any shade cloth. If these trees grew in a mature forest, they would have been below the canopy, competing for light and other resources and thus might not have responded as much as we observed in our competition-limited conditions (Polgar & Primack, 2011). However, our results may have implications beyond how mature forests function.

Our focus on juvenile trees may suggest how forests regenerate and assemble with the longer, warmer seasons of anthropogenic climate change. Plant phenology research has suggested that earlier seasons with warming favor the species that shift earlier the most, with many studies focused on invasive species appearing to support this (Wolkovich *et al.*, 2013; Zettlemoyer *et al.*, 2019). Yet, these studies are often on herbaceous species, and our results, along with other recent work (Zani *et al.*, 2020; Zohner *et al.*, 2023), suggest mid- and later-season effects may have underappreciated performance implications in woody species. Thus, more studies of how later-season events correlate with invasions, and forest assembly more generally, could be especially important for understanding how warming will shift forest regeneration.

Our results also suggest that population-level differences in assembly models may not be as critical as previously suggested (Cavender-Bares, 2019). While studies of large latitudinal gradients have found lower growth in tree species from poleward latitudes (Aitken & Bemmels, 2016; Soolanayakanahally *et al.*, 2013), we found little difference in growth across a latitudinal gradient of 3.5°. Thus, at least across this latitudinal range, local adaptation on growth for these species appears weak. Supporting this, we found no evidence of shifting end of season across provenances in these four closely related species, on average, despite decades of work finding that end of season events, such as budset, are usually more locally adapted than start of season events such as leafout (Aitken & Bemmels, 2016; Habjørg, 1978; Keller *et al.*, 2011; Kramer, 1936; Soolanayakanahally *et al.*, 2013; Zeng & Wolkovich, 2024). Yet, a study of the same common garden, including all tree and shrub species found end of season appeared to vary strongly by year and shift earlier with earlier start of season (Buonaiuto *et al.*, 2026), suggesting end of season may be more variable than currently proposed (Alberto *et al.*, 2013). However, compared to other studies that found strong local adaptation on growth and budset (e.g. Rohde *et al.*, 2011), we had little replication per species across a smaller latitudinal gradient. Taken together, these results imply that our current understanding of budset being strongly locally adapted and thus varying little with temperature or start of season needs more study.

1.4.1 Implications for large-scale vegetation modeling

Using a rarely available dataset of tree growth coupled with leaf phenology start and end season, we found an overall positive response from multiple tree species to longer seasons, which has important implications for models of future carbon storage. While juvenile trees store little carbon compared to their adult forms, they are crucial for forest regeneration capacity and range expansion (as discussed above, and see Luu *et al.*, 2024), and generally predict adult tree responses qualitatively (Wolkovich *et al.*, 2025). We found that the start and end of season asymmetrically matter to growth, which means that how we model these phenological events in vegetation and related earth system models may be critical for accurate predictions. Modelling the end of season better will require more studies, especially on more species across a larger gradient. This could help tease out whether local adaptation of end of season depends on the species, latitudinal range (both distance and how far north provenances come from), or some combination of the two, which could then feed into improved models.

Integrating our results into models will also require considering how environmental conditions beyond temperature shift across the season, especially soil moisture (Gao *et al.*, 2022; Li *et al.*, 2023). Trees in our common garden were regularly watered, likely relieving any major drought pressures, though studies beyond ours have found a connection between a later the end of season and more growth, (see: Gao *et al.*, 2022; Wang *et al.*, 2025). For the four species we studied, there are currently no increases in drought occurrence within their ranges (Agel *et al.*, 2025; National Centers for Environmental Information, 2026), suggesting that their growth may increase with longer and warmer growing seasons (Agel *et al.*, 2025; Cook & Pederson, 2011), as we found. However, the positive effect of a later end of season could be offset if soil moisture becomes inadequate for growth, and other temperate regions with decreased soil moisture have reported declines in tree growth (e.g., Chiang *et al.*, 2021; Spinoni *et al.*, 2018). Improved accuracy of future carbon sequestration models would consequently require integrating over more regions and growth drivers.

Chapter 2

Weak response of warmer growing seasons on tree growth across 11 species of adult trees in an urban arboretum

2.1 Introduction

Anthropogenic climate change affects many different biological processes with one of the most documented being shifts in phenology—the timing of recurring life history events (Cleland *et al.*, 2007; Menzel *et al.*, 2006; Parmesan *et al.*, 1999). Earlier springs and later autumns have lengthened the window of opportunity during which plants can grow (Duputié *et al.*, 2015).

For trees, longer growing seasons mean they should grow more. This is especially significant given the critical role that forests play in offsetting human-emitted greenhouse gases (Harris *et al.*, 2021; Keenan *et al.*, 2014; Stridbeck *et al.*, 2022). In temperate forests, the largest 1% of trees store half of the living biomass, which is largely composed of carbon sequestered from the atmosphere (Lutz *et al.*, 2018; Mensah & Petersson, 2024). Given that temperate forests account for 40% of the global carbon sink (Gibbs *et al.*, 2025), this makes mature trees disproportionately important in global carbon dynamics. The future of this major carbon sink, however, largely relies on whether these trees will keep growing with longer growing seasons.

The long standing assumption that longer growing seasons should increase tree growth has recently been challenged by a suite of studies that failed to find this relationship (Cabon *et al.*, 2022; Charlet De Sauvage *et al.*, 2022; Dow *et al.*, 2022; Silvestro *et al.*, 2023; Čufar *et al.*, 2015). Recent research suggests that warmer springs, which predominantly drive extended growing seasons, have not increased the maximum tree growth rate or annual radial growth in temperate mixed forest communities (Dow *et al.*, 2022). Similarly, Čufar *et al.* (2015) showed no shift in annual radial growth in relation to the growing season length for *Fagus Sylvatica*. However, neither of these studies could clearly establish why they observed this decoupling, leaving open the question of why trees may not grow more with longer growing seasons.

Tree growth strategies may shape how different species respond to extended growing seasons. Longer growing seasons may benefit slower growing species because appropriate conditions would allow them to keep their low growth rate active for a longer period (Cuny *et al.*, 2012). In contrast, faster growing species may complete their annual growth more quickly, making extra growing days less important (Cuny *et al.*, 2012). However, there is a possibility that since faster-growing species are generally more flexible and responsive, they could opportunistically take advantage of changing conditions more than more conservative species (Baumgarten *et al.*, 2026).

Wood anatomy in angiosperm trees may additionally impact when they grow, with implications for their responses to longer seasons. Ring-porous species have large xylem vessels that allow them to source water deep into the ground, but also make them more prone to vessel damage from drought or spring frost (Kitin & Funada, 2016; Taneda & Sperry, 2008). This damage means that they generally start radial growth several days or weeks before they leafout, as they need to restore their hydraulic system before new leaves can develop and unfurl (Barbaroux & Breda, 2002; D'Orangeville *et al.*, 2022; Hessl *et al.*, 2026; Savage *et al.*, 2022). In contrast, diffuse-porous species have smaller xylem vessels, and thus source comparatively shallower water but with a lower risk of vessel damage (Yin *et al.*, 2023). Because of this, diffuse-porous species usually leafout earlier than ring-porous species (because they do not need to restore their water transportation pathway), but generally start and stop radial growth after ring-porous species (D'Orangeville *et al.*, 2022; Michelot *et al.*, 2012; Yin *et al.*, 2023). Additionally, these differences mean that leafout and the start of radial growth are often synchronized in diffuse-porous species, making this leaf phenophase a good proxy for growth, while earlier events (e.g., budburst) may be more relevant for ring-porous species (Čufar *et al.*, 2008).

The differing timing of growth for species with different wood anatomy means they experience considerably different temperature regimes during the growing season. This may result in an asymmetric effect of season length because trees that start growing early experience cooler conditions than species that start growing later. As ring-porous species start growing earlier—when temperatures are cooler—and complete their growth earlier in the season (D'Orangeville *et al.*, 2022; Michelot *et al.*, 2012), they may experience a cooler overall growing season than diffuse-porous species. Given the importance of temperature to growth rates (Chuine & Régnière, 2017; Wang & Engel, 1998), we may thus expect that diffuse-porous species will grow more with longer and warmer seasons, as much of their growth will occur at higher temperatures with a corresponding higher growth rate. Diffuse-porous species could also grow more with longer seasons because they may have faster life-history strategies that allow them to rapidly take advantage of longer seasons compared to ring-porous species. There is diverging evidence, however, of a direct correlation between wood porosity and life-history strategies; Lv *et al.* (2025) recently proposed that diffuse-porous species are slower-growing and longer-living than ring-porous species, in contrast to previous work that proposed the opposite (Taneda & Sperry, 2008). Beyond wood anatomy, however, species may also vary in how they allocate carbon to growth versus other functions.

How trees store the carbon they accumulate—whether it is through an immediate investment in growth or for long-term storage—could mute short-term responses to longer growing seasons (Hessl *et al.*, 2026). Investing in longer-term storage could mean that the previous year's thermal conditions may affect growth in the following year. Deciduous trees in particular often rely on stored carbon at the beginning of the growing season to grow new leaves before starting photosynthesis (Monson *et al.*, 2018; Wolkovich *et al.*, 2025; Zweifel *et al.*, 2006). Mature trees may even rely on carbon that was stored for several years for their early-season growth (Hessl *et al.*, 2026). Thus, while environmental conditions during the current year should directly influence growth, carbon allocation strategies may blur annual growth responses.

Here we examine variation in how the growing season affects growth across 11 species in an urban arboretum. We consider both the calendar growing season (in days) and, given the importance of temperature to growth rates, the thermal season (summed daily temperatures across the calendar season). Leveraging data from a citizen science monitoring program, we estimate these two measures of growing season length from leafout to leaf coloring across nine years. Our data include 49 mature deciduous angiosperm trees with varying wood anatomies and growth rates. Thus, we ask how the current and previous years' growing seasons affect annual radial growth across a suite of species.

2.2 Methods

2.2.1 Citizen science program

The Tree Spotters is a citizen science program started in 2015 that aims to train citizen scientists for accurate and rigorous phenological monitoring at the Arnold Arboretum of Harvard University (42.30 °N, -71.12 °W; Figure 2.1). From 2015 to 2024, hundreds of citizen scientists monitored 77 individual trees and shrubs of 14 species. They regularly followed individuals from budburst in the spring to leaf coloring in the fall using the National Phenology Network (NPN) phenophases (Denny *et al.*, 2014): leaves (483), colored leaves (498), fruits (516), ripe fruits (390), falling leaves (471), recent fruit or seed drop (504), increasing leaf size (467), breaking leaf buds (371), flowers or flower buds (500), open flowers (501), pollen release (502). Not all phenophases were recorded for every tree, for every year, and some trees are missing several years of data.

Here we use leafout to colored leaves (or leaf coloring) to define the growing season length. Taken together, our dataset comprises 321 leafout and 284 colored leaves observations (which we refer to as the 'full datasets'), but we did not have both observations for all trees in all years, so our total of matched full growing seasons duration (which we refer to as the 'restricted dataset') is 278 observations. We visually checked for leafout, colored leaves and growing season length outliers and verified that there was no observation bias potentially caused by a particular observer or in a particular year.

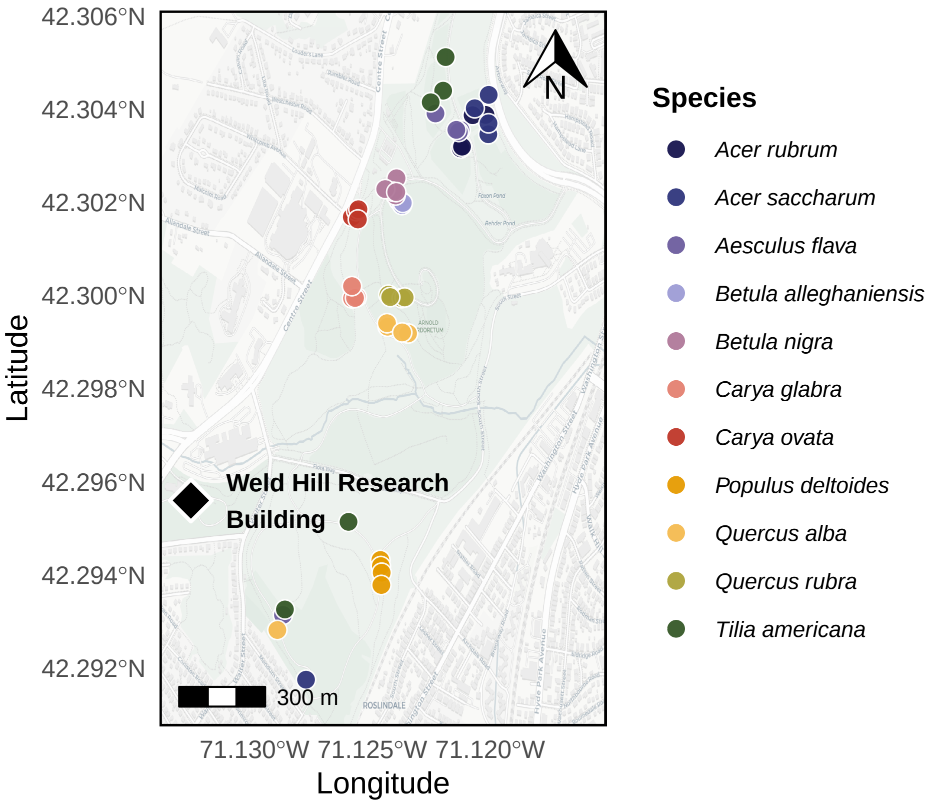

Figure 2.1: Map of the Arnold Arboretum of Harvard University with the locations of the trees depicted as dots. The colours of the dots match the model estimates of model estimate figures (e.g. Figure 2.2). We also show the location of the Weld Hill Research Building (black diamond), where the common garden from Chapter 1 took place.

2.2.2 Phenological monitoring and sample collection

From 20 to 22 April 2025, we collected tree ring samples only for the tree species that were previously monitored for phenology—excluding *Fagus grandifolia* because of beech bark disease concerns—for a total of 49 individuals (Table 2.1). Thus, we sampled four or five individuals for 11 species: red maple (*Acer rubrum*), sugar maple (*Acer saccharum*), yellow buckeye (*Aesculus flava*), yellow birch (*Betula alleghaniensis*), river birch (*Betula nigra*), pignut hickory (*Carya glabra*), shagbark hickory (*Carya ovata*), eastern cottonwood (*Populus deltoides*), white oak (*Quercus alba*), northern red oak (*Quercus rubra*) and american basswood (*Tilia americana*). For this, we collected two 5-mm diameter cores, 15 cm long at 1.3 m above ground, perpendicular to the slope and at 180 degrees from each other, using an increment borer (Mora Borer, Haglöfs Sweden, Bromma, Stockholm, Sweden). We cleaned the increment borer with alcohol (70% ethanol) and the inside with a brush before collecting each core. We stored the cores in modified plastic straws to promote aeration and left them to dry at ambient temperature for three months.

2.2.3 Sample processing, imaging and measuring

We used wooden mounts to stabilize the cores that we then sanded using progressively finer sandpaper grit (from 150 to 1000) by increments of roughly 150. We then used a high-resolution scanner to scan the cores at a resolution of 6250 dpi (Fong *et al.*, 2026) and used those images to measure the ring widths using ImageJ (Schneider *et al.*, 2012).

We visually cross-dated the ring width chronologies by assigning a calendar year to each ring and comparing annual variation of trends across the different individuals of the same species (Cook & Kairiukstis, 1990). Our short chronologies and low species replication prevented us from performing statistical cross-dating. We also did not detrend our chronologies because the ring widths during our study period (nine years) showed no consistent allometric trend (Figure A.2). We included year in our models to control for the effects of different years.

2.2.4 Data analyses

We used Bayesian hierarchical models to understand how growth shifted with four different season metrics: GDD, GSL, SOS and EOS. We used growing degree days (GDD; mean daily temperature above 5°C) between leafout and leaf coloring to calculate the thermal season and the number of days between leafout and leaf coloring for the calendar season (growing season length; GSL). Then, we used leafout for the start of season (SOS) and leaf coloring for the end of season (EOS). To better illustrate ring width responses to our season metrics, we report ring width change (log(mm)) in response to the number of GDDs accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD) for the thermal season. Similarly, we report ring width change per week of season length, leafout and leaf coloring for GSL, SOS and EOS, respectively. We report effects when the 90% UI do not overlap with zero across all models.

We combined daily climate data from two sources located next to the Arboretum for the duration of our study (2016-2024). From 2016-2019, we used the Weld Hill research station climate station (Arnold Arboretum, 2026) and the "Boston (Weld Hill)" station for the remaining years (2020-2024; Carroll *et al.*, 2026). We determined drought occurrence at the arboretum using the North American Drought Monitor Maps and used the proposed drought index to qualify their intensity (National Centers for Environmental Information, 2026).

Thus, for each season predictor, we estimated its effect on ring width (i, denoting each observation) observed for each tree (*treeid*) in each year (*year*) as:

$$
\begin{aligned}
ln(ringwidth)_i &\sim Normal(\mu, \sigma_y) \\
\mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\
\alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha\,\text{treeid}})
\end{aligned}
\tag{2.1}
$$

With this model, we estimated a species-specific effect of each season predictor on growth ($\beta_{species}$), but we also controlled for additional variation due to species ($\alpha_{species}$), individual tree (α_{treeid}) and year (α_{year}). We ran the models using both natural units and scaled predictors, using z-scoring, which allows us to directly compare effect sizes of different predictors (Gelman & Hill, 2006). We report estimates as means and 90% uncertainty intervals (UI). We also fit the models for SOS and EOS using the full vs restricted datasets. Finally, because the wood growth onset is synchronized with a different leaf phenophase for species with different wood anatomy (ring-porous being more closely synchronized with budburst and diffuse-porous, with leafout), we ran the models using budburst instead of leafout as the SOS. This allowed us to test whether our model responses diverged across diffuse vs ring porous species (Table 2.1).

We used an additional model to account for the previous year's thermal conditions. For this, we used the GDD accumulated between leafout and leaf coloring from the previous year (e.g. for the growth year of 2020, we used the GDD of 2019 as the previous year's GDD), alongside the GDD accumulated during the current year. For interpretability, we also report ring width change (log(mm)) as the number of GDDs (for both years) accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD). Thus, we estimated the previous and current year thermal season effect on the current year's growth as:

$$
\begin{aligned}
\ln(\text{ringwidth})_i &\sim Normal(\mu, \sigma) \\
\mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} \\
&\quad + \beta^{\text{current}}_{\text{species}} \times \text{GDD}_{\text{current}} + \beta^{\text{previous}}_{\text{species}} \times \text{GDD}_{\text{previous}} \\
\alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha_{\text{treeid}}})
\end{aligned}
\tag{2.2}
$$

While controlling for the same variation across species, individual tree and year, this model allowed us to account for the effect of the thermal season during the current year ($\beta^{\text{current}}_{\text{species}}$), but also during the previous year ($\beta^{\text{previous}}_{\text{species}}$).

Finally, we used a third Bayesian hierarchical model to determine whether the SOS and EOS of our species were related during the current year. For this, we estimated the effect of the SOS on the EOS on each tree, in each year as:

$$
\begin{aligned}
\mathrm{EOS}_i &\sim Normal(\mu_i, \sigma_y) \\
\mu_i &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{\text{species}} \times \mathrm{SOS} \\
\alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha_{\text{treeid}}})
\end{aligned}
\tag{2.3}
$$

2.3 Results

Over the duration of the study, mean summer temperatures spanned 21.53°C (2017) to 23.07°C (2020), with an average of 22.51°C. 2020 and 2022 had moderate droughts in the summer (June, July and August), which ramped up to severe in September. Additionally, a moderate drought in June 2016 ramped up to severe in July, then extreme in August and September.

The calendar season (GSL) ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021), while the thermal season (GDD) ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD (*T. americana* in 2016). The earliest start of season (leafout) was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020), while the earliest end of season (leaf coloring) day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016). Annual ring widths across years varied from 0.43 mm (*A. saccharum* in 2019) to 11.75 mm (*T. americana* in 2019; Figure A.14).

Across our eleven species and season predictors, we did not find any clear response of growing season length on growth. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth 'RW/day' or 'RW/GDD' with days and GDD being adjusted to represent more interpretable units (see 'Data analyses' section in Methods for more details on predictor adjustment). A longer thermal season did not increase growth for any species and decreased growth for *B. alleghaniensis* (β = -0.08, UI: -0.12, -0.03 RW/GDD; Figure 2.2a and e; Table A.7). Further, a longer calendar season did not change growth for any species (Figure 2.2b and f; Table A.7). On a scaled axis (through z-scoring, Gelman & Hill, 2006), the thermal and calendar seasons were similarly predictive (see Figure A.22).

Previous year' climate also had limited effects on growth. For the one species—*B. alleghaniensis*—that showed decreased growth with warmer thermal seasons, this effect persisted for the current year, but previous year's GDD actually increased growth ($\beta = 0.15$, UI: 0.02, 0.28 RW/GDD; Figure 2.3b; Table A.9). The congeneric of *B. alleghaniensis*, *B. nigra*, in contrast grew less with previous year's GDD (β = -0.08, UI: -0.17, -0.00 RW/GDD; Figure 2.3b; Table A.9) and current year GDD increased growth ($\beta = 0.12$, UI: 0.03, 0.21 RW/GDD; Figure 2.3a; Table A.9). Interestingly, *B. nigra* did not solely respond to the current year's conditions (Figure 2.2a and e; Table A.7), but grew more with warmer thermal seasons only when the previous conditions were considered.

An earlier SOS did not increase growth for any species and decreased growth for *T. americana* ($\beta = 0.15$, UI: 0.02, 0.27 RW/day; Figure 2.2c and g; Table A.7), which did not respond to the thermal or the calendar season. Additionally, a later end of season did not increase growth for any species and decreased growth for *B. alleghaniensis* (β = -0.09, UI: -0.16, -0.01 RW/day; Figure 2.2d and h; Table A.7), which also decreased growth with warmer thermal seasons. Estimates were unchanged when using the full or restricted datasets (SOS: $n = 321$ and EOS: $n = 284$ vs $n = 278$; Figure A.20b), or when using budburst instead of leafout as the start of the growing season (Figure A.21c).

Finally, a later SOS delayed EOS only for *P. deltoides* (β: 1.04, UI: 0.39, 1.70 EOS/SOS; Figure A.23; Table A.10). We did not find any clear difference in baseline growth between species (Figure A.17; A.18) and across year s (Figure A.13; Table A.8), even in 2016, where there was an extreme drought (α: -0.21, UI: -0.77, 0.34 RW; Figure A.13; Table A.8). Growth was similar across individual trees (Figure A.19).

Table 2.1: Species studied here grouped by growth speed and wood anatomy. We collected the growth speed information for each species from Burns *et al.* (1990) and the wood anatomy from different sources (Cheng *et al.*, 2024; D'Orangeville *et al.*, 2022; Grover *et al.*, 2023). We sampled a total of 49 individual trees

Latin binomial (Common Name)	Growth speed	Wood anatomy	n
Acer rubrum (red maple)	Moderate	Diffuse-porous	4
Acer saccharum (sugar maple)	Slow	Diffuse-porous	5
Aesculus flava (yellow buckeye)	Moderate	Diffuse-porous	5
Betula alleghaniensis (yellow birch)	Moderate	Diffuse-porous	4
Betula nigra (river birch)	Fast	Diffuse-porous	4
Carya glabra (pignut hickory)	Slow	Ring-porous	4
Carya ovata (shagbark hickory)	Moderate	Ring-porous	5
Populus deltoides (eastern cottonwood)	Fast	Diffuse-porous	4
Quercus alba (white oak)	Slow	Ring-porous	5
Quercus rubra (northern red oak)	Moderate	Ring-porous	4
Tilia americana (american basswood)	Fast	Diffuse-porous	5

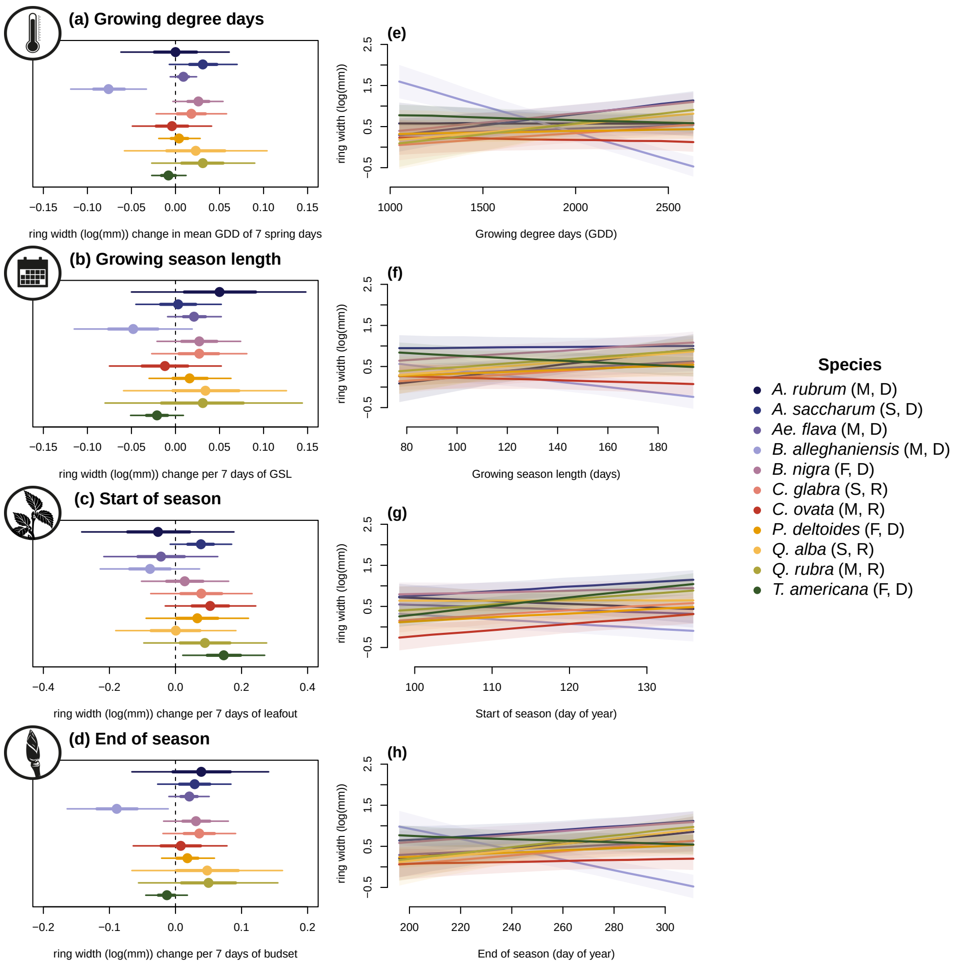

Figure 2.2: Estimates of ring width change (log(mm)) as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length; start of season; end of season across 11 species: (red maple (*Acer rubrum*), sugar maple (*Acer saccharum*), yellow buckeye (*Aesculus flava*), yellow birch (*Betula alleghaniensis*), river birch (*Betula nigra*), pignut hickory (*Carya glabra*), shagbark hickory (*Carya ovata*), eastern cottonwood (*Populus deltoides*), white oak (*Quercus alba*), northern red oak (*Quercus rubra*) and american basswood (*Tilia americana*)); species colors are the same across the panels). In the legend, the growth speed of each species is depicted as slow (S), moderate (M) and fast (F) and ring-porous (R) and diffuse-porous (D) correspond to their wood anatomy (Table 2.1). Panels on the left (a to d) represent the mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor, which we represent through ring width change in mm on the log scale per adjusted predictor (see 'Data analyses' section in Methods for more details on predictor adjustment). See Table A.7 for parameter estimates. Panels on the right (e to h) show the ring width response curves as a function of each predictor, with the lines depicting the mean response at each predictor value, and the shaded area, the 50% uncertainty intervals. These models do not include the effect of the previous year's GDD (see Figure 2.3 for these estimates).

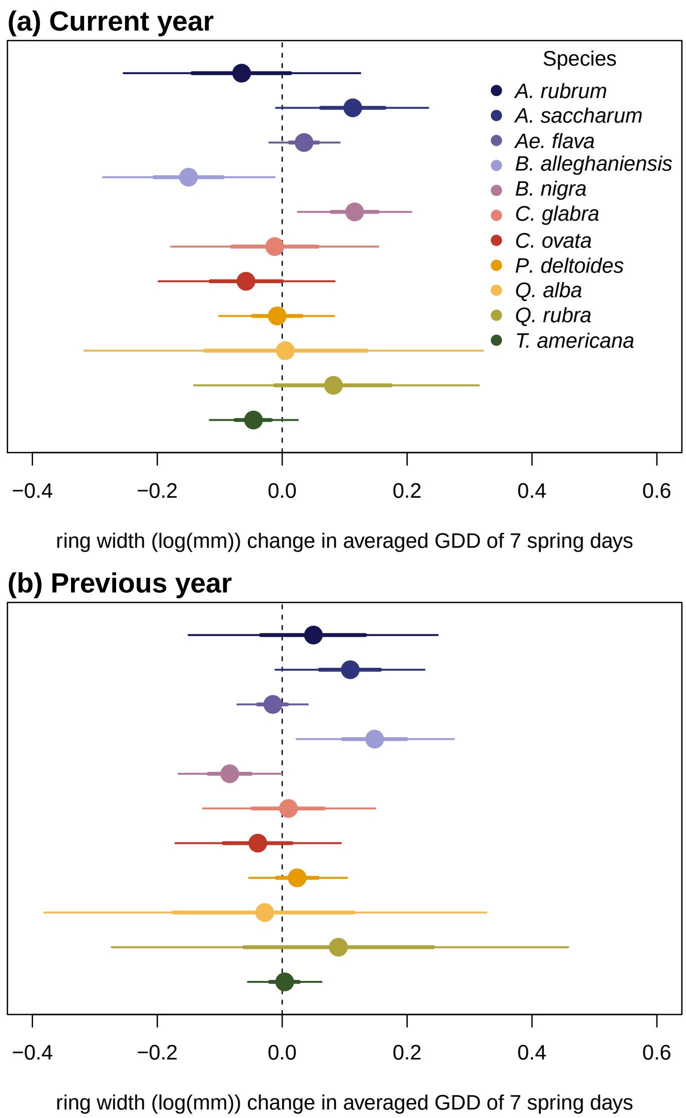

Figure 2.3: Model slope estimates representing ring width change (log(mm)) as a function of the current year growing degree days (GDD; a) and the previous year's GDD (b) from a model considering both effects. GDD represent the accumulated growing degree days (mean daily temperature above 5°C) between leafout and leaf coloring, and the models' estimates show ring width change (log(mm)) per scaled GDD (see 'Data analyses' section in Methods). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

2.4 Discussion

In this study, we used 49 trees of 11 species to understand if mature trees grew more with longer growing seasons. To do this, we used annual growth measured from tree rings and leaf phenology to estimate the start and end of season. Our findings reveal that warmer and longer growing seasons did not increase growth for any species and decreased growth for one. We also found that the previous year's thermal conditions shifted growth trends for two species.

Our findings support recent results that trees do not grow more with longer growing seasons. Some studies that have failed to find a connection between growth and growing season length did not consider species differences (e.g., Cabon *et al.*, 2020), but others have looked specifically across species. The most comparable study (with eight species in common with our dataset; Dow *et al.*, 2022) was conducted geographically close to ours and found similar results: despite warmer and earlier springs, trees did not grow more. Our results complement these findings as we used leaf phenology and have the end and start of season, while the previous study only had the start of season for growth phenology and not leaf phenology. In addition, Čufar *et al.* (2015) had substantially more years (five decades) and found analogous results to ours, but only for one European species, *Fagus Sylvatica*. Taken together, these results support the hypothesis that longer growing seasons may not increase temperate tree growth, at least for mature trees.

We previously showed (see Chapter 1) that juvenile trees in a common garden grew more with longer and warmer growing seasons, contrasting with the response we found here for mature trees. Interestingly, these diverging responses support the assumption that younger trees are usually more responsive to changing conditions compared to mature trees (Wolkovich *et al.*, 2025). The common garden study looked at fewer species (only four) with rather homogeneous life-history strategies (all fast growing), but our study on mature trees in the arboretum comprised more species (11) spanning a wider range of life-history strategies. Despite this larger sample of species, mature trees did not show a strong signal of life-history strategies in their growth response to growing season conditions, with even the fastest growing species (i.e. *B. nigra*, *P. deltoides* and *T. americana*) barely showing a response to warmer seasons.

2.4.1 Life-history influence on growth trends

While our species share an array of growth strategies (see Table 2.1), we did not find a consistent growth trend across them. Only *B. alleghaniensis*, which has a moderately fast growth, responded to the thermal season, and its decrease in growth suggests it may be negatively impacted by warmer seasons. *B. alleghaniensis* also grew less with a later end of season, which is a period characterized by fast GDD accumulation. The fact that *B. alleghaniensis* appears to be negatively impacted by warmer seasons suggests that this species could face declines at its southernmost range, where its growth may not keep up with warmer and longer seasons (Aitken *et al.*, 2008; Doak & Morris, 2010). In contrast to *B. alleghaniensis*, its congeneric species, *B. nigra*, increased in growth with warmer seasons (though only when the previous year's conditions were included). *B. nigra* is faster growing than *B. alleghaniensis*, which may explain its response, though we did not find any consistent response across the other fast-growing species. Indeed, the fast-growing *T. americana* (Table 2.1; Burns *et al.*, 1990) grew less with an earlier start of season—when the days are short and cool. This is in contrast with our expectation of species with this rapid life-history strategy, which should provide an edge by starting to grow earlier in the season (Loughnan *et al.*, 2025). The other eight species had different life-history strategies (fast: *P. deltoides*; moderate: *A. rubrum*, *A. flava*, *C. ovata*, *Q. rubra*; slow: *A. saccharum*, *C. glabra*, *Q. alba*; Table 2.1), and despite this diversity in growth speed, none changed their growth with longer seasons. Ultimately, these findings suggest a weak signal of life-history strategies in growth responses to longer and warmer growing seasons.

We also found no clear trends with ring porosity. *B. alleghaniensis* and *B. nigra*, which showed the contrasting responses described above, were both diffuse-porous and the five other diffuse-porous species (*A. flava*, *A. rubrum*, *A. saccharum*, *P. deltoides* and *T. americana*) did not increase growth with longer or warmer seasons. We had only four ring-porous species (*C. glabra*, *C. ovata*, *Q. alba* and *Q. rubra*), which responded similarly to the diffuse-porous species. While this is a small sample, it supports recent work (e.g., Grover *et al.*, 2023) suggesting ring porosity may not be as useful a predictor for growth as previously suggested (Wang *et al.*, 1992). While wood porosity imposes inherent physiological differences across species, this dichotomous classification may oversimplify the reality of the complex dynamic of tree growth. In fact, a gradient of life history strategies may exist within each wood porosity type where both fast- and slow-growing species can share the same wood porosity, though this is poorly documented (Taneda & Sperry, 2008). This emphasizes the need to better classify growth traits and relate carbon allocation to wood porosity.

2.4.2 Growth response may be blurred by carbon allocation to storage

Shifts in our findings when including the previous year's thermal season suggest that including carbon allocation strategies may be important to understanding how longer growing seasons affect growth. The growth response of two *Betula* species shifted when we included the previous year's thermal season as a driver. These two congeneric and diffuse-porous species (*B. alleghaniensis* and *B. nigra*) had diametrically opposed responses, however. *B. alleghaniensis* growth decreased with the current year's thermal conditions, but increased with a warmer previous year, possibly revealing a carbon allocation strategy that prioritizes carbon storage above short-term growth. This is in contrast with the expectation that diffuse-porous species will tend to allocate less carbon to storage and more to growth (Barbaroux & Breda, 2002; Muhr *et al.*, 2016). In contrast, *B. nigra* increased in growth during the current year—only when the previous year's conditions were included—supporting the expectation that diffuse-porous species prioritize rapid investment in growth with less allocation to storage. These opposite responses of two closely related species highlight the importance of considering more than the current year's environmental conditions in tree growth and growing season length models.

Taken together, our results suggest that understanding how tree growth shifts with longer seasons will need more theory and study of what species differences drive diverging responses (Delpierre *et al.*, 2016; Silvestro *et al.*, 2023; Čufar *et al.*, 2008). Broad categorical definitions of growth strategies (e.g., fast or slow) and wood anatomy may be too simple for the complexity observed across species. Further, allocation to other processes could make the effect of longer growing seasons dependent on both past environmental conditions and investments by plants in previous years to storage or reproduction. Since most trees have a two-year or longer cycle from bud to seed, including reproductive investment may improve models. If carbon allocation to storage and reproduction, life-history strategies and differences in wood anatomy blur growth responses as we suggest, further studies leveraging a larger sample size—both more species observed over more years—should be specifically designed to tease these factors apart.

Chapter 3

Conclusions and future work

We used observational data to understand how longer and warmer seasons change growth for juvenile and mature trees (Figure 3.1), with some species in both studies that showed no response. Our findings are significant because they suggest that temperate forest regeneration may benefit from longer and warmer seasons, which could further be beneficial for range expansion. Interestingly, and in contrast to our expectation, provenance did not influence growth, contrasting previous work on the importance of local adaptation on end of season phenology and growth (Aitken & Bemmels, 2016; Soolanayakanahally *et al.*, 2013; Zeng & Wolkovich, 2024). In addition, mature trees store most of forests' biomass, but in our study they failed to grow more with longer growing seasons. One species was negatively affected by warmer seasons, and the others did not change their growth. Our results hint that this absence of trend could be the result of poorly understood multi-year lag effects of carbon allocation to storage instead of growth. Taken together, these findings reveal how the complex nature of climate change, when coupled with the complicated tree growth dynamics, make drawing general conclusions across species hard. These confounding factors may limit the interpretation of current observational data studies, but they also suggest a pathway for improved studies.

Ideal observational studies and how experiments could tease apart confounding factors

Future observational studies including a higher number of species would be useful to address many of questions my thesis raised, but how to select those species is particularly important. Because evolutionary history can alter a suite of traits for species (Pagel, 1999), future observational studies may want to leverage this to better test specific traits. For example, studies should consider having a minimum of two congeneric species that have distinct life-history strategies to reflect the importance of these strategies while correcting for certain traits, such as wood anatomy, that arise from shared evolutionary history. One relevant example for northeastern North American forests (as I studied here) would be to simultaneously study *Acer negundo* (fast-growing and short-

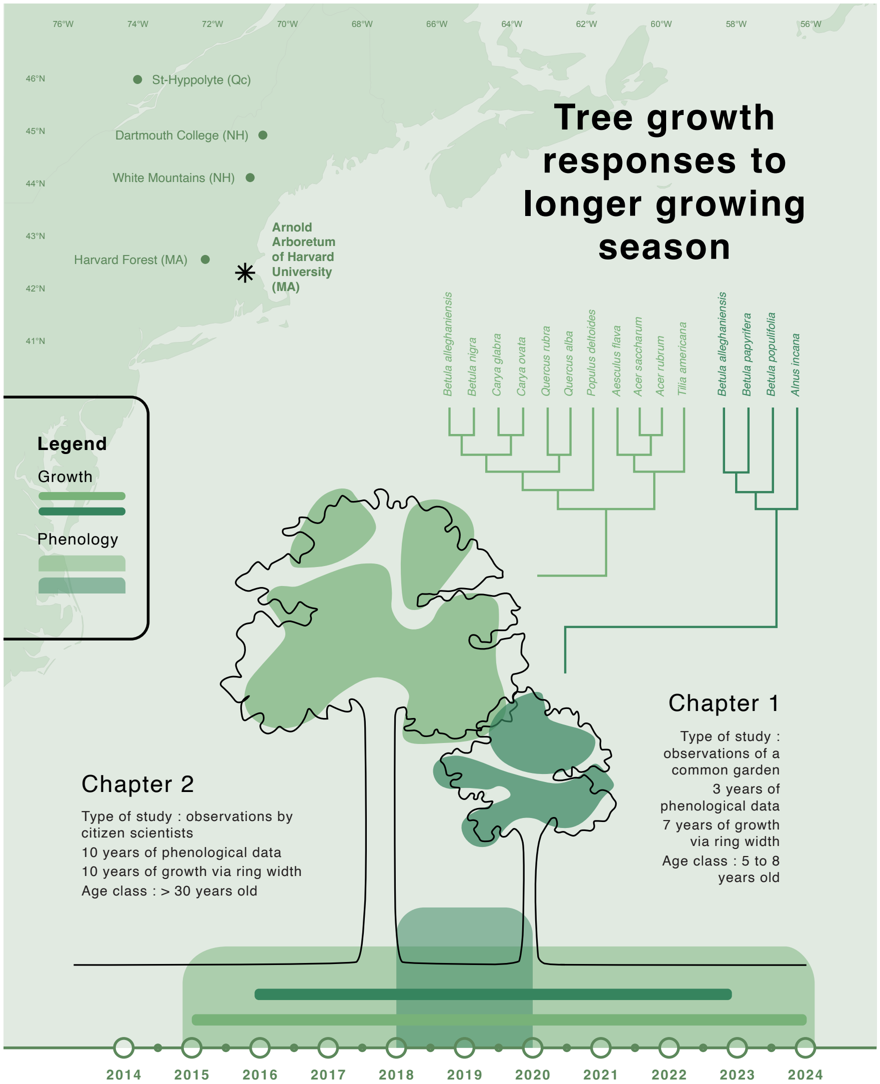

Figure 3.1: Overview of the age class, species, provenance of the trees used in each study along with the type of study each chapter consists of. The colored lines represent the period of the growth data and the shaded areas are the phenological data. Both are colour-coded by project. Each phylogram illustrates the species used in each project. The map illustrates the four different provenances used in Chapter 1, and the star represents the study location of both chapters

living) and *Acer saccharum* (slow-growing and long-living; Burns *et al.*, 1990). Further, studying these species for a longer period that spans different life stages (seedling to adult) would also provide useful information on their growth plasticity across their lifetime—a response we could not well establish with our studies. Finally, leaf phenology data provides some information on the growth timing of trees, but tools such as dendrometers, could further increase our understanding of intra-annual growth dynamics and confirm whether primary and secondary growth are coupled. These approaches would also help establish whether our conclusion that the end of season is more important than the start of season for juvenile trees was appropriate. While such studies would require deploying intensive resources, experiments can also provide useful inference on some of these drivers.

Logistical constraints of working with adult trees mean that experiments are most often performed on juvenile trees. However, even if young trees are often more plastic than adult forms, their responses can still provide valuable insights into differences across species (Vitasse, 2013; Wolkovich *et al.*, 2025). This is why experiments that are specifically designed to tease apart confounding factors that are inevitable in observational studies (such as the effect of spring vs autumn warming) would be useful to complement our studies. In addition, these experiments should be conducted across an array of species with a gradient of life-history strategies so we can increase our understanding of the mechanisms regulating growth speed with changing conditions.

While conducting the analyses of my two chapters, I also investigated the impact of a growing season extension and different stressors on growth (Figure 3.2) under experimental conditions. First, I participated in an experiment that looked at the effects of drought and defoliation (mimicking insect outbreaks) intensity and timing on the phenology and growth of six tree species (Figure 3.2: Phaenoflex). This experiment will provide valuable insights into these constraints on growth that are likely to increase in frequency with climate change. In parallel to my two chapters, I also conducted a large-scale experiment in which we subjected seven species of juvenile trees to spring and autumn treatments in a full factorial design of Warm/Cool, Spring/Autumn (Figure 3.2: Fuelinex). This experiment will be particularly helpful in testing the validity of our conclusions from the observational studies. If these trends contrast, the full factorial design of this experiment will help us understand the importance of shifts in spring versus autumn phenology on growth. I ran this experiment for two years and have completed biomass sampling, but I will finalize allometry and modeling work from the study later; Fuelinex will become the first chapter of my PhD thesis that will start in January 2027.

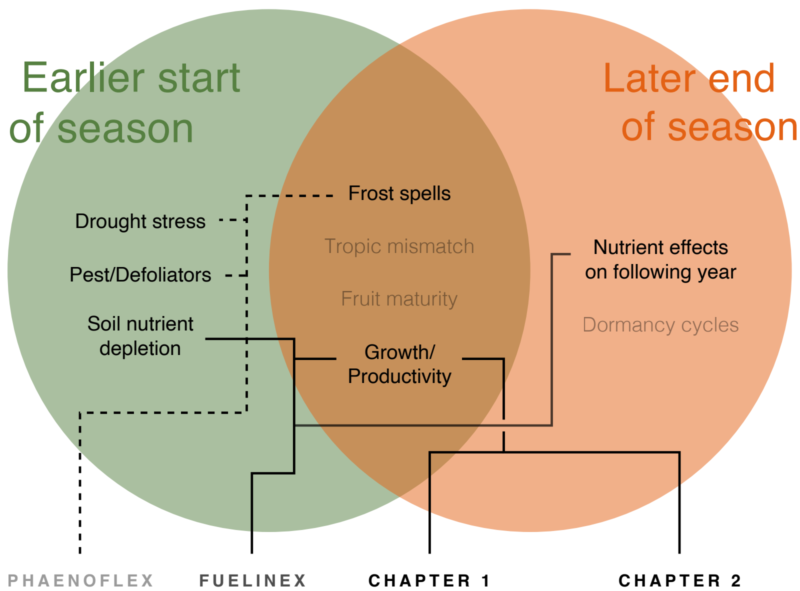

Figure 3.2: The effects that an earlier start and later end of season can have on trees. In addition to my thesis work, Fuelinex is an experiment that I conducted over the last two and a half years, and that will become my first PhD chapter. Solid lines connect the effects that I studied over the course of this thesis (Chapter 1 and 2 and Fuelinex). Phaenoflex (in grey) and its dashed lines represent other effects I investigated in a related experimental project that I collaborated on in 2023 and 2024.

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Appendix A

Supplemental Material

Table A.1: Number of individuals sampled per species and per provenance for a total of 81 individual trees sampled from the United States of America (in Massachusetts (MA) and New Hampshire (NH)) and Canada (in Quebec (Qc)).

Provenance	*A. incana* ($n = 29$)	*B. alleghaniensis* ($n = 20$)	*B. papyrifera* ($n = 7$)	*B. populifolia* ($n = 25$)
Harvard Forest (MA)	7	0	4	7
White Mountains (NH)	6	7	0	9
Dartmouth College (NH)	8	7	2	9
St-Hippolyte (Qc)	8	6	1	0

Table A.2: Provenance coordinates and 1951–1980 climate normals: mean annual temperature (MAT; °C), mean summer precipitation (MSP; mm), and frost-free period (FFP; days). We extracted the climate variables at each provenance from (Wang *et al.*, 2016)

Provenance	Latitude	Longitude	MAT	MSP	FFP
Harvard Forest (MA)	42.5	-72.2	7.4	450	133
White Mountains (NH)	44.1	-71.0	6.8	434	128
Dartmouth College (NH)	44.9	-70.7	3.2	490	110
St-Hippolyte (Qc)	46.0	-74.0	3.4	472	121

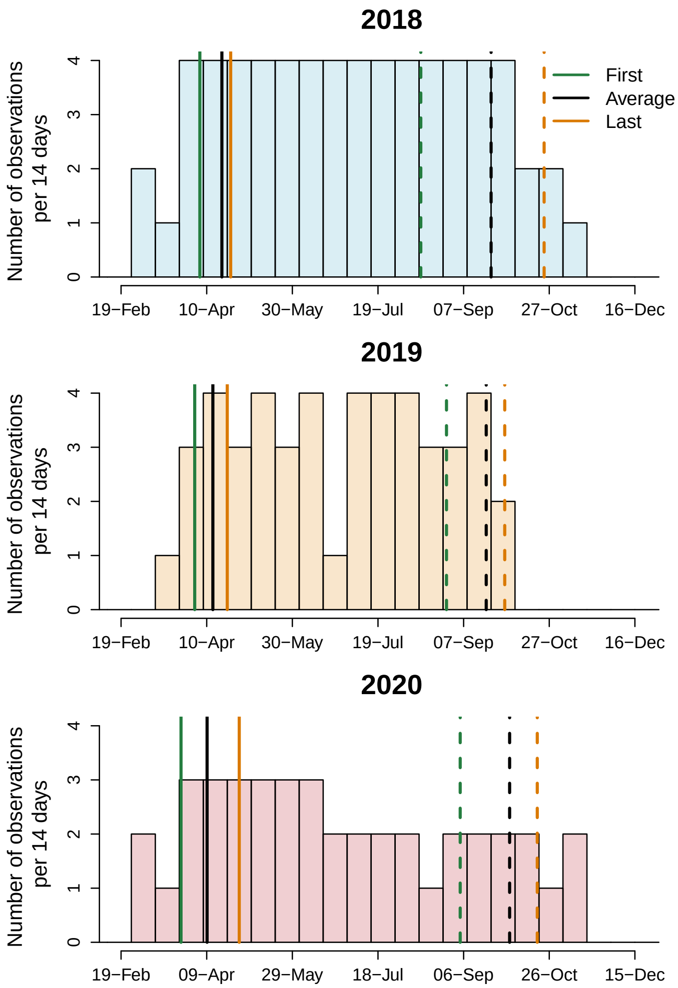

Figure A.1: Leaf phenology monitoring frequency across years. We monitored leaf phenology once or twice per week for three years, though the frequency varied within and across years. Each histogram bin represents the number of observations that were made during a period of 14 days and the solid lines represent the earliest, the average and last recorded budburst observations; the dashed lines depict the budset observations. This demonstrates that across the different frequencies of observation, distances between early spring phenophases were similar across years.

Table A.3: Species-level slope parameters which represent ring width change (log(mm)), per adjusted predictor (see 'Data analyses' in Methods for more details). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different phenological predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Species	GDD	GSL	SOS	EOS
A. incana	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.01, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.15 [0.06, 0.23]	0.18 [0.05, 0.32]	0.05 [-0.11, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.07 [0.02, 0.12]	0.06 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.03, 0.17]

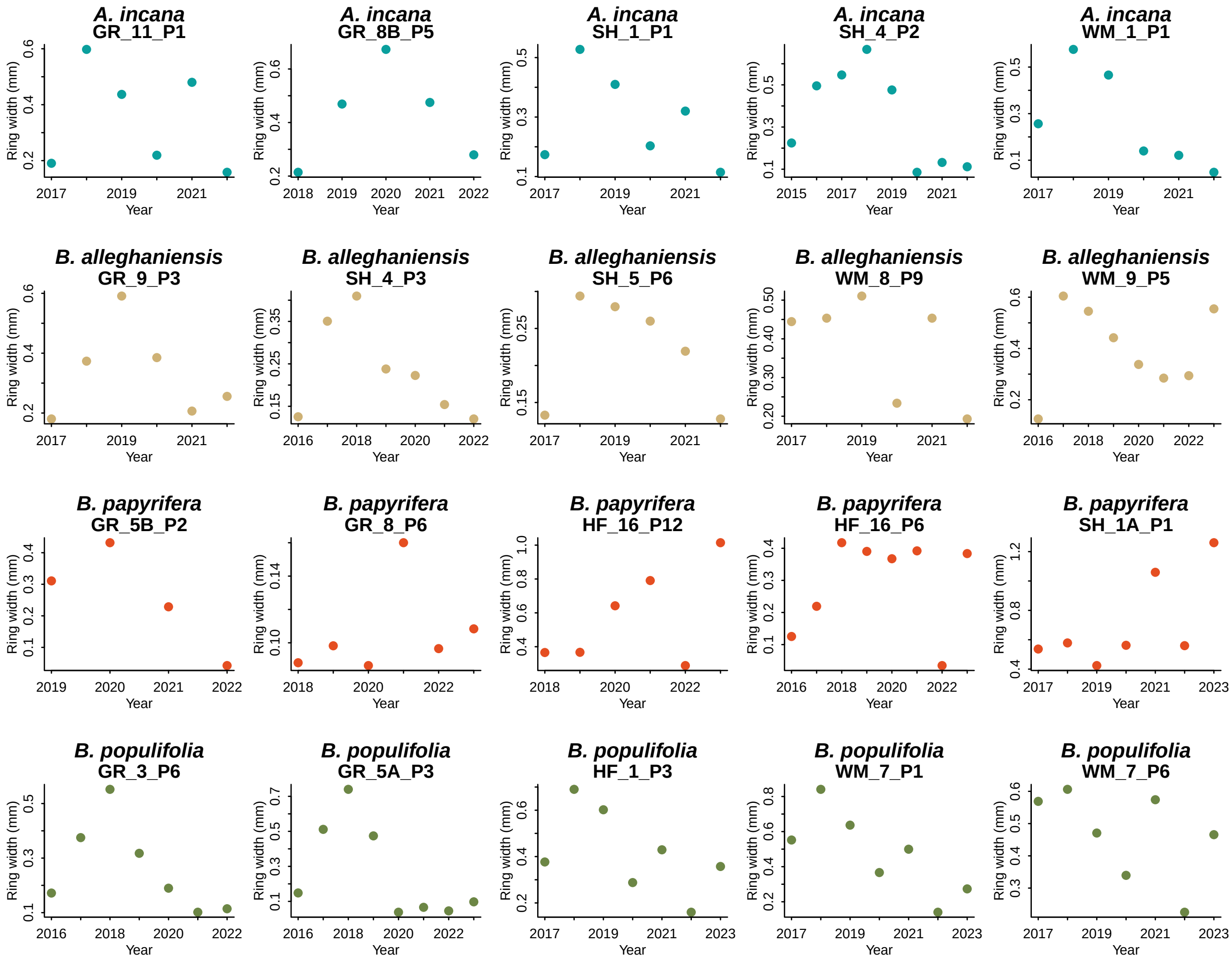

Figure A.2: We show the potential allometry patterns across all years we had ring width for. For this, we randomly sampled 20 trees from our dataset and represented their ring width (mm) for each year.

Table A.4: The effect of provenance on ring widths (log(mm)). We report these intercept estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Provenance	GDD	GSL	SOS	EOS
Harvard Forest (MA)	-0.10 [-0.33, 0.07]	-0.08 [-0.32, 0.09]	-0.05 [-0.25, 0.12]	-0.09 [-0.33, 0.08]
White Mountains (NH)	0.08 [-0.09, 0.29]	0.08 [-0.08, 0.31]	0.09 [-0.07, 0.30]	0.08 [-0.09, 0.29]
Dartmouth College (NH)	-0.03 [-0.22, 0.14]	-0.04 [-0.23, 0.14]	-0.06 [-0.26, 0.10]	-0.03 [-0.23, 0.15]
St-Hippolyte (Qc)	0.04 [-0.14, 0.24]	0.04 [-0.14, 0.26]	0.02 [-0.15, 0.22]	0.04 [-0.14, 0.26]

Table A.5: Summary output of each parameter from our four predictors, using ring width (log(mm)) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See 'Data analyses' in Methods section for more details.

Parameter	GDD	GSL	SOS	EOS
α	-0.62 [-4.69, 3.53]	0.04 [-3.90, 4.04]	1.12 [-2.61, 4.85]	-1.33 [-5.48, 2.73]
$\sigma_{\alpha_{treeid}}$	0.17 [0.05, 0.27]	0.17 [0.05, 0.28]	0.21 [0.11, 0.30]	0.18 [0.06, 0.28]
$\sigma_{\alpha_{prov}}$	0.17 [0.02, 0.47]	0.17 [0.02, 0.48]	0.17 [0.02, 0.46]	0.18 [0.02, 0.49]
σ_y	0.48 [0.43, 0.53]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{species_{A.incana}}$	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{species_{B.alleghaniensis}}$	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.01, 0.20]	0.12 [0.05, 0.18]
$\beta_{species_{B.papyrifera}}$	0.15 [0.06, 0.23]	0.18 [0.05, 0.32]	0.05 [-0.11, 0.21]	0.21 [0.09, 0.33]
$\beta_{species_{B.populifolia}}$	0.07 [0.02, 0.12]	0.06 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.03, 0.17]
$\alpha_{prov_{DartmouthCollege(NH)}}$	-0.03 [-0.22, 0.14]	-0.04 [-0.23, 0.14]	-0.06 [-0.26, 0.10]	-0.03 [-0.23, 0.15]
$\alpha_{prov_{HarvardForest(MA)}}$	-0.10 [-0.33, 0.07]	-0.08 [-0.32, 0.09]	-0.05 [-0.25, 0.12]	-0.09 [-0.33, 0.08]
$\alpha_{prov_{St-Hippolyte(Qc)}}$	0.04 [-0.14, 0.24]	0.04 [-0.14, 0.26]	0.02 [-0.15, 0.22]	0.04 [-0.14, 0.26]
$\alpha_{prov_{WhiteMountains(NH)}}$	0.08 [-0.09, 0.29]	0.08 [-0.08, 0.31]	0.09 [-0.07, 0.30]	0.08 [-0.09, 0.29]
$\alpha_{year_{2018}}$	0.19 [-0.77, 1.15]	0.22 [-0.71, 1.19]	0.21 [-0.73, 1.15]	0.24 [-0.72, 1.17]
$\alpha_{year_{2019}}$	0.18 [-0.79, 1.14]	0.09 [-0.85, 1.06]	0.11 [-0.83, 1.05]	0.12 [-0.84, 1.06]
$\alpha_{year_{2020}}$	-0.43 [-1.40, 0.53]	-0.35 [-1.29, 0.61]	-0.38 [-1.32, 0.58]	-0.47 [-1.44, 0.47]
$\alpha_{species_{A.incana}}$	1.06 [-3.19, 5.21]	0.20 [-3.87, 4.27]	1.54 [-2.12, 5.13]	3.08 [-1.30, 7.45]
$\alpha_{species_{B.alleghaniensis}}$	0.21 [-3.97, 4.34]	0.44 [-3.65, 4.46]	-1.89 [-5.68, 1.93]	-1.81 [-6.32, 2.74]
$\alpha_{species_{B.papyrifera}}$	-4.00 [-8.89, 0.75]	-2.60 [-7.15, 1.93]	-0.77 [-5.06, 3.47]	-5.54 [-11.15, 0.12]
$\alpha_{species_{B.populifolia}}$	-0.66 [-4.94, 3.63]	0.28 [-3.96, 4.44]	-0.48 [-4.35, 3.35]	-0.88 [-5.32, 3.75]

Table A.6: Summary output of each parameter from our four predictors, using basal area increment ($\log(mm^2)$) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See 'Data analyses' section for more details.

Parameter	Estimate
α	0.62 [-4.52, 5.84]
$\sigma_{\alpha_{treeid}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{prov}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{species_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{species_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{species_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{species_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{species_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{species_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{species_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{species_{B.populifolia}}$	-0.07 [-5.98, 5.57]

Table A.7: Species-level slope parameters which represent ring width change (log(mm)), per scaled predictor (see 'Data analyses' section in Methods for more details on predictor scaling). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). Table A.8 reports all the model's parameters.

Species	GDD	GSL	SOS	EOS
A. rubrum	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
A. saccharum	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
Ae. flava	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
B. alleghaniensis	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
B. nigra	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
C. glabra	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
C. ovata	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
P. deltoides	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
Q. alba	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
Q. rubra	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
T. americana	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

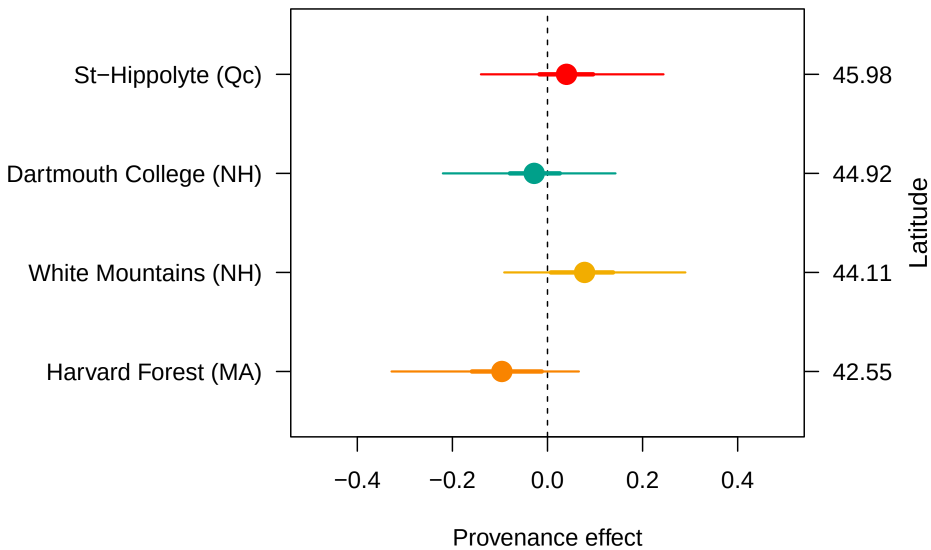

Figure A.3: The effect of provenance on growth across a latitudinal gradient. We show the ring width (log(mm); see Table A.8 for all parameter estimates) for each provenance with growing degree days as the predictor (effects were similar for the other predictors; see Table A.4). The dots represent the estimated mean of each provenance, with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively). The provenances are color-coded from Figure 2.1, the map showing their locations. We also performed a separate analyses, fitting the provenances per species, individually (see Figure A.4)

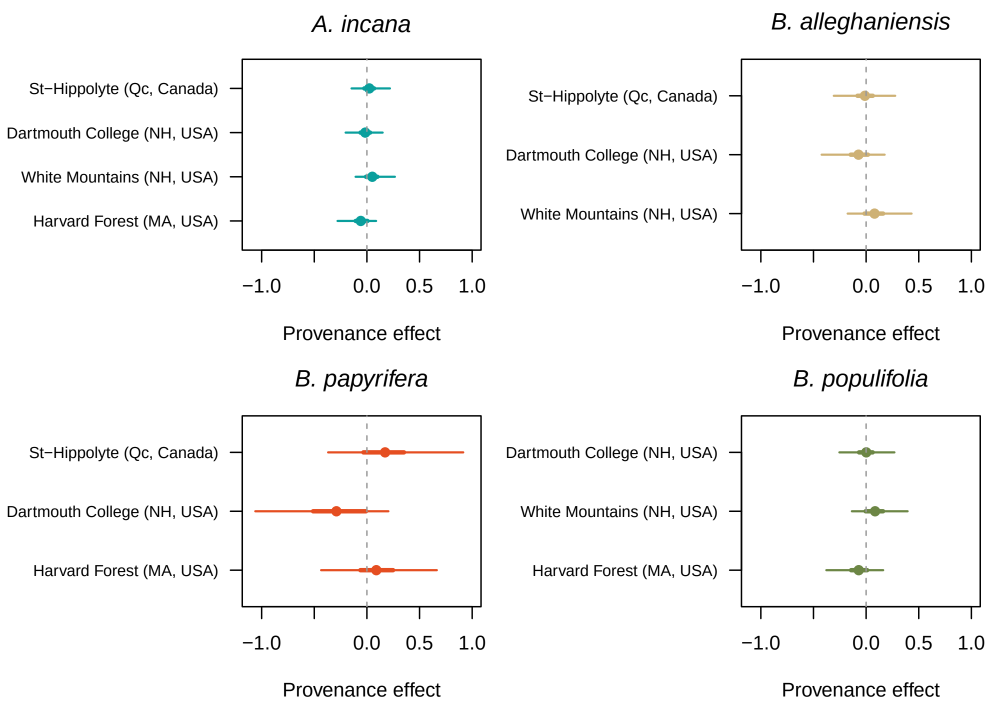

Figure A.4: The effect of provenance on growth across a latitudinal gradient estimated per species. We show the ring width (log(mm)) for each provenance estimated from the following intercept-only model: $log(ringwidth) \sim Normal(\alpha_{provenance} + \alpha_{treeid}, \sigma)$, using the package rstanarm version 2.32.1 (Gabry & Goodrich, 2016). The dots represent the estimated mean of each provenance (ordered by latitude), with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively).

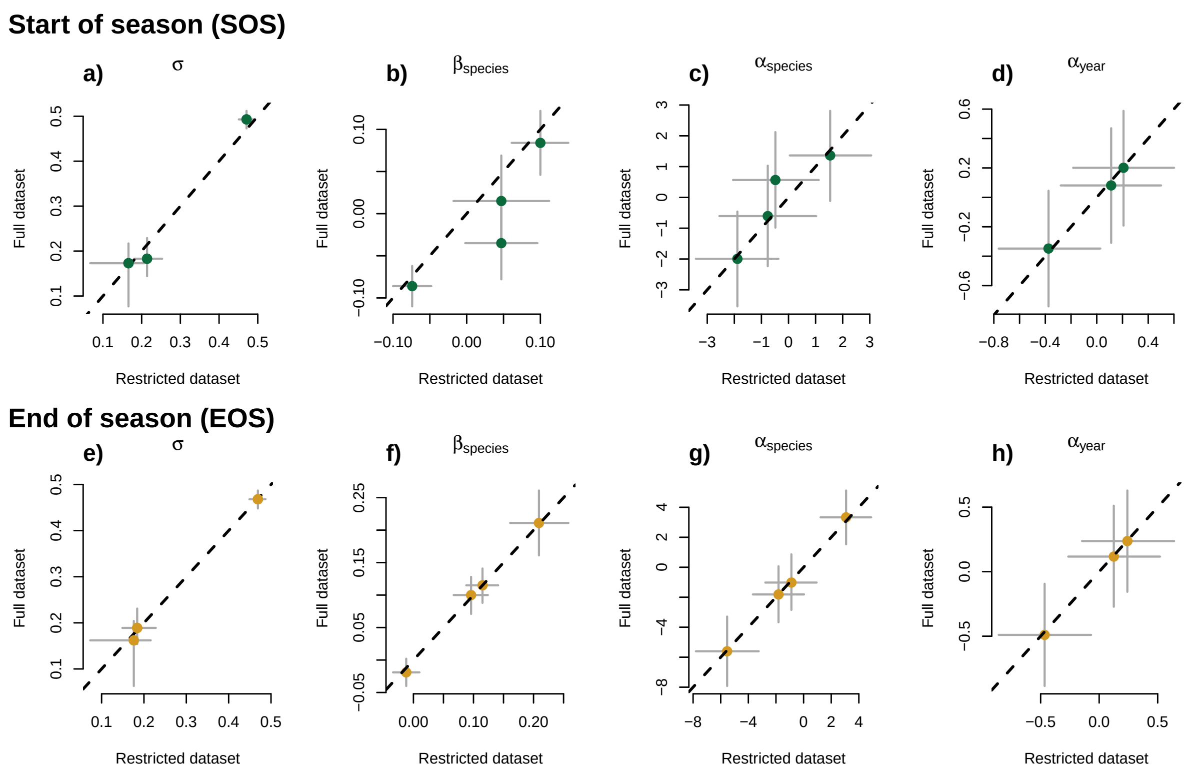

Figure A.5: Full vs most restricted datasets for the model with start of season (a-d) and end of season (d-h) as the predictor. Each panel represent the different parameters used to fit the models. σ (a, e), $\alpha_{species}$ (c, g) and α_{year} (d,h) represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ (b, f) represents the change in ring width (log(mm))/adjusted predictor, for each species (see 'Data analyses' section in Methods for more details).

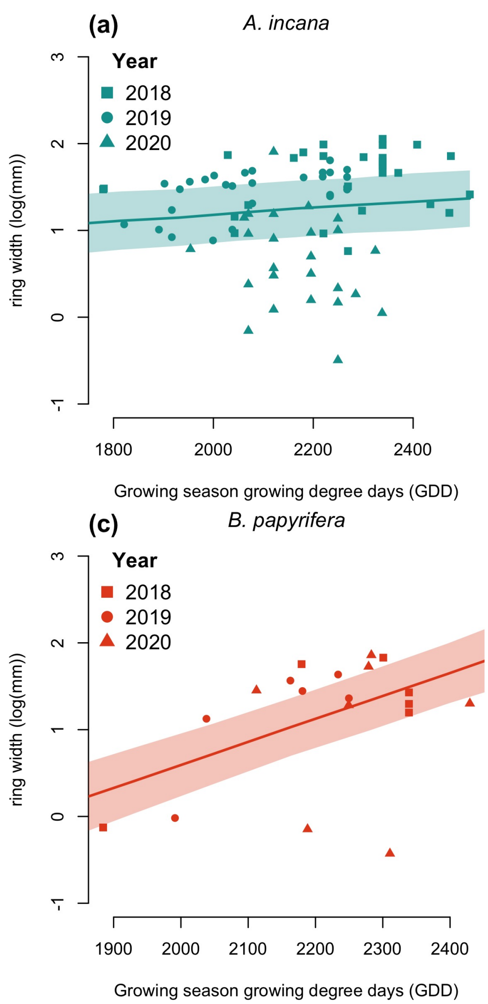

Figure A.6: Ring width (log(mm)) trends in response to thermal season (growing degree days). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI at each predictor value. The dots are the empirical data shaped by year.

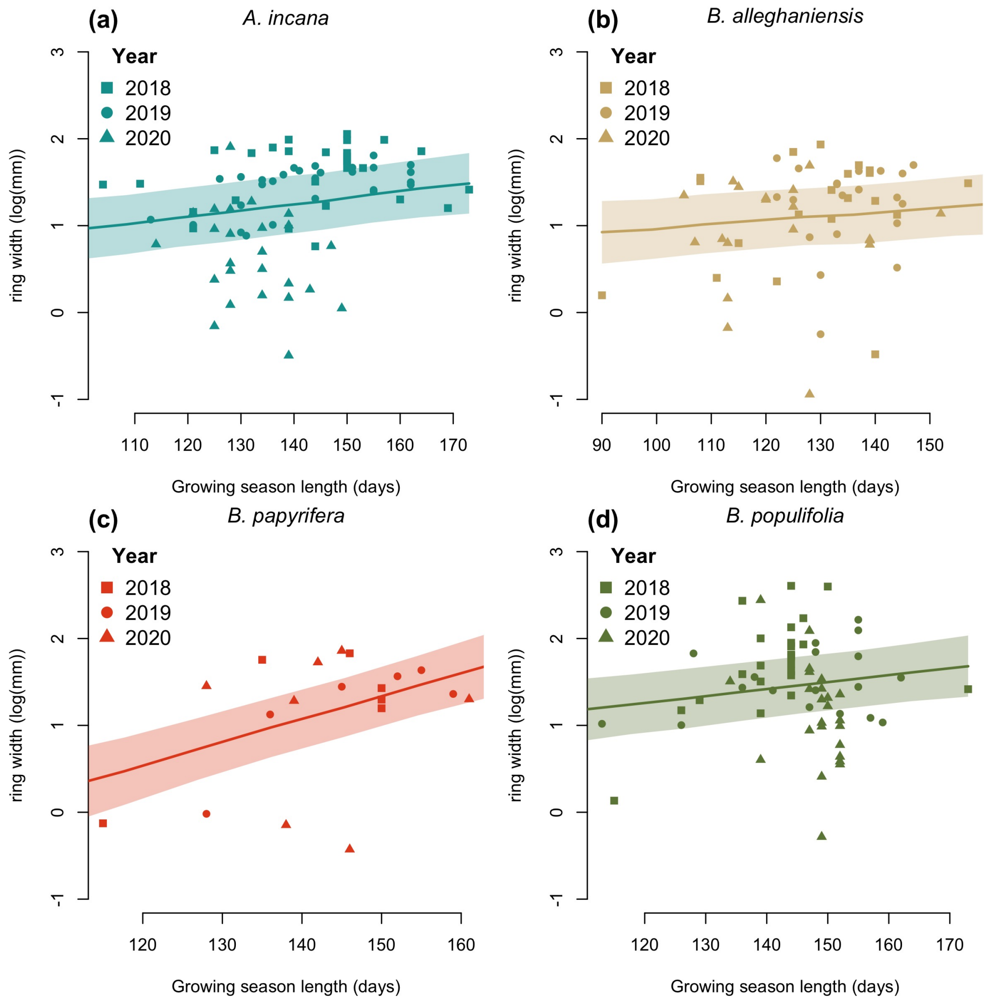

Figure A.7: Ring width trends (log(mm)) in response to calendar season (growing season length). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI at each predictor value. The dots are the empirical data shaped by year.

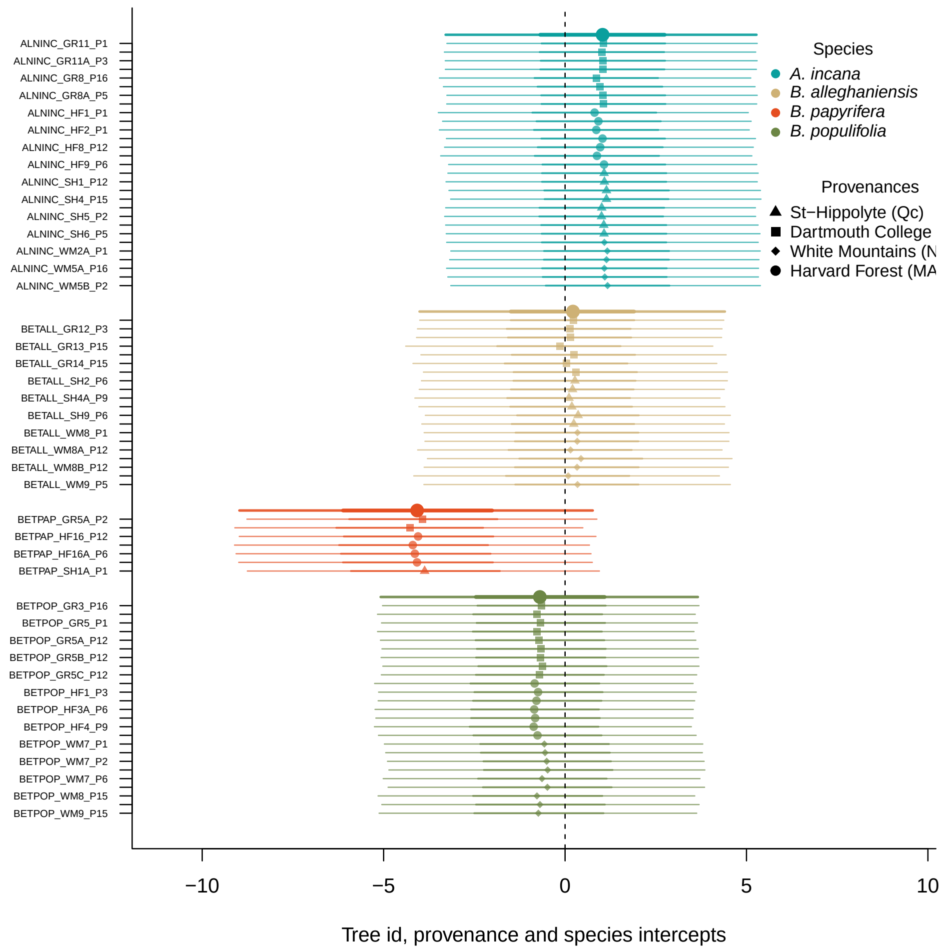

Figure A.8: Ring width (log(mm)) deviations from the grand mean for grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species-level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution, and the thick and thin lines, the 50 and 90% UI.

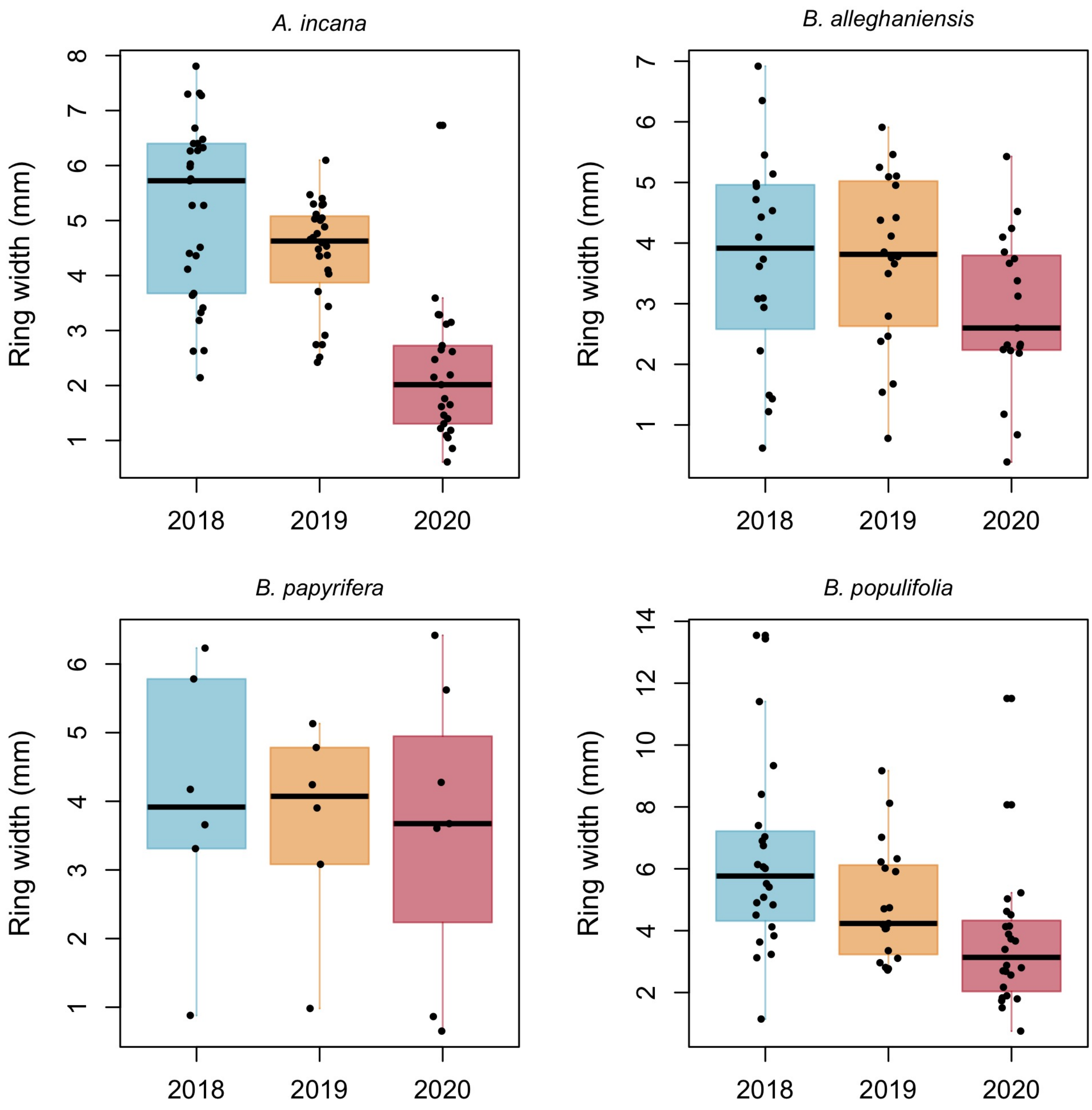

Figure A.9: Boxplots representing the empirical data of our ring widths (mm) separated by species. The dots represent the raw data, the line is the mean and the box depicts the 50% UI.

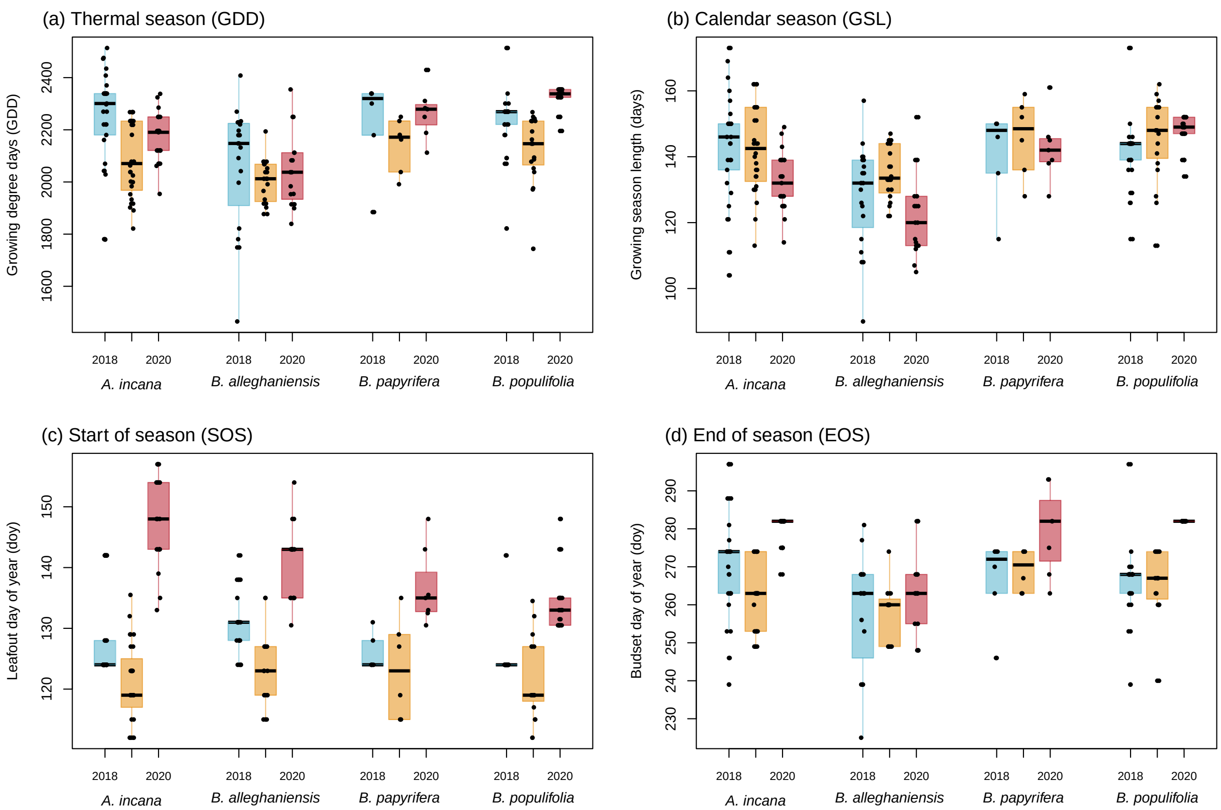

Figure A.10: Boxplots representing the empirical data of all our predictors clustered by species, and for each year. The dots represent the raw data, the line is the mean and the box depicts the 50% UI.

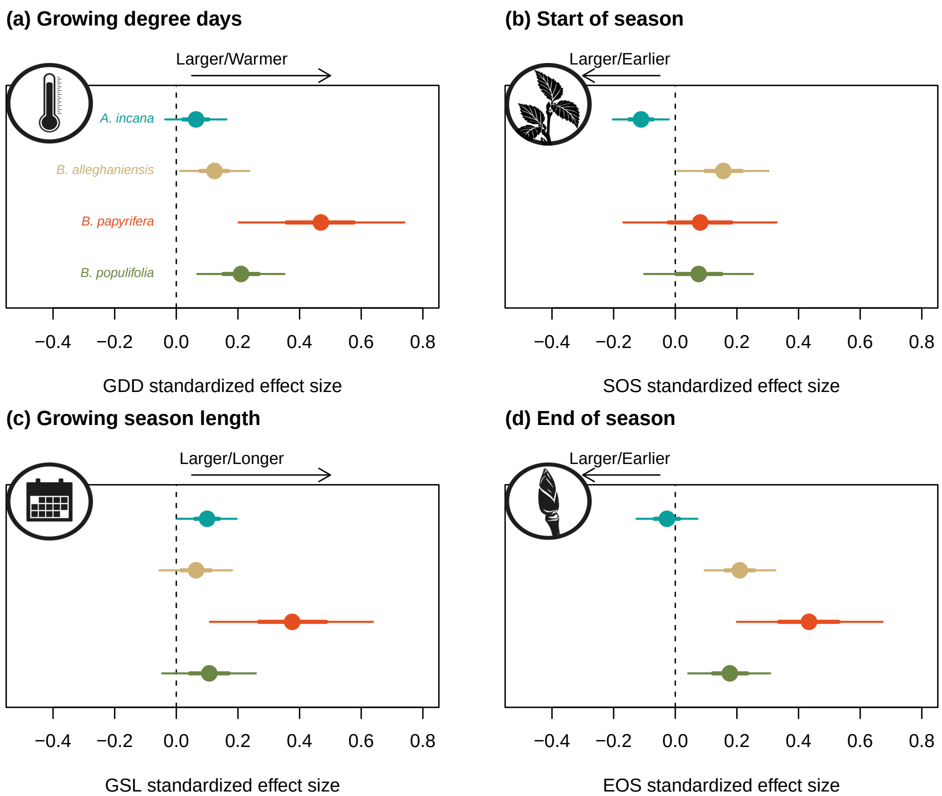

Figure A.11: Effect size of each predictor across our four species. We depict ring width (log(mm)) change per adjusted predictor (z-scored; see 'Data analyses' section for more details) for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (growing degree days, growing season length, start of season and end of season).

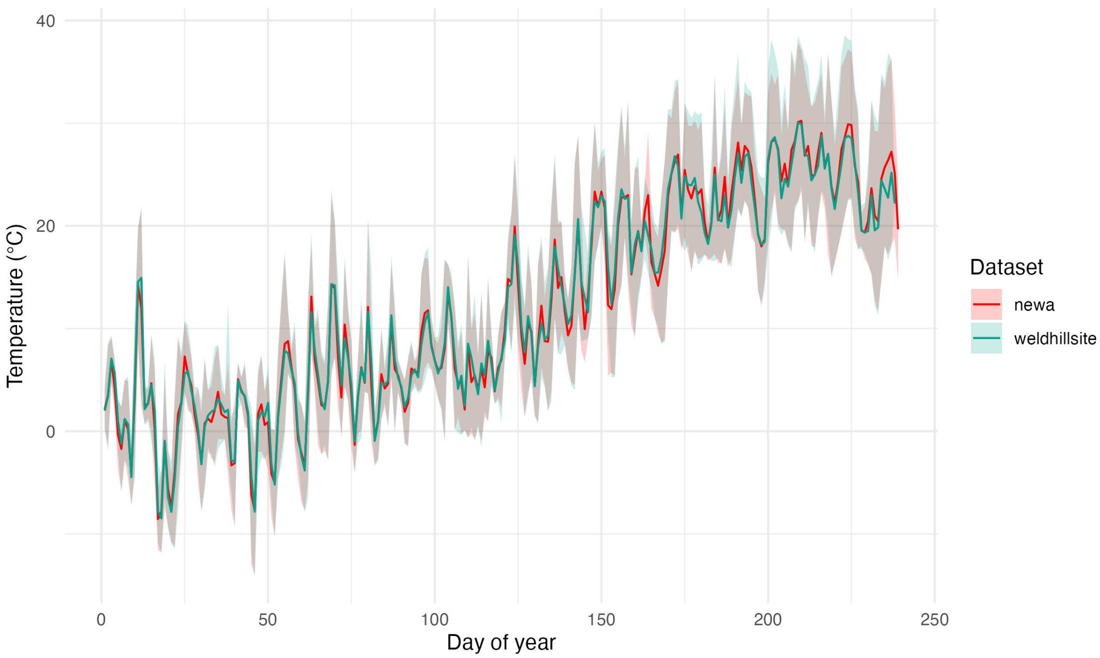

Figure A.12: Climate data overlap in 2020 for the two climate stations we used in our dataset (see 'Data analyses' section for more details). The lines represent the daily mean temperature of each climate station and the shaded area, the minimum and maximum daily temperatures.

Table A.8: Summary output of each parameter from our four predictors, using ring width (log(mm)) as the response variable (See equation 2.1). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See 'Data analyses' section in Methods for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{species_{A.rubrum}}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{species_{A.saccharum}}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{species_{Ae.flava}}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{species_{B.alleghaniensis}}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{species_{B.nigra}}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{species_{C.glabra}}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{species_{C.ovata}}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{species_{P.deltoides}}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{species_{Q.alba}}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{species_{Q.rubra}}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{species_{T.americana}}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year_{2016}}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{year_{2017}}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{year_{2018}}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{year_{2019}}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{year_{2020}}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{year_{2021}}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{year_{2022}}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{year_{2023}}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{year_{2024}}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{species_{A.rubrum}}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{species_{A.saccharum}}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]
$\alpha_{species_{Ae.flava}}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{species_{B.alleghaniensis}}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{species_{B.nigra}}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{species_{C.glabra}}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{species_{C.ovata}}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{species_{P.deltoides}}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{species_{Q.alba}}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{species_{Q.rubra}}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{species_{T.americana}}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]

Table A.9: Summary output of each parameter from the previous year model (Equation 2.2) using ring width (log(mm)) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets). see 'Data analyses' section in Methods for more details

Parameter	Estimate
α	0.20 [-1.10, 1.48]
$\sigma_{\alpha_{tree}}$	0.58 [0.47, 0.72]
σ_y	0.29 [0.26, 0.32]
$\beta^{\text{current}}_{\text{species}_{A.rubrum}}$	-0.07 [-0.25, 0.12]
$\beta^{\text{current}}_{\text{species}_{A.saccharum}}$	0.11 [-0.01, 0.23]
$\beta^{\text{current}}_{\text{species}_{Ae.flava}}$	0.04 [-0.02, 0.09]
$\beta^{\text{current}}_{\text{species}_{B.alleghaniensis}}$	-0.15 [-0.29, -0.01]
$\beta^{\text{current}}_{\text{species}_{B.nigra}}$	0.12 [0.03, 0.21]
$\beta^{\text{current}}_{\text{species}_{C.glabra}}$	-0.01 [-0.18, 0.15]
$\beta^{\text{current}}_{\text{species}_{C.ovata}}$	-0.06 [-0.20, 0.08]
$\beta^{\text{current}}_{\text{species}_{P.deltoides}}$	-0.01 [-0.10, 0.08]
$\beta^{\text{current}}_{\text{species}_{Q.alba}}$	0.01 [-0.32, 0.32]
$\beta^{\text{current}}_{\text{species}_{Q.rubra}}$	0.08 [-0.14, 0.32]
$\beta^{\text{current}}_{\text{species}_{T.americana}}$	-0.05 [-0.12, 0.03]
$\beta^{\text{previous}}_{\text{species}_{A.rubrum}}$	0.05 [-0.15, 0.25]
$\beta^{\text{previous}}_{\text{species}_{A.saccharum}}$	0.11 [-0.01, 0.23]
$\beta^{\text{previous}}_{\text{species}_{Ae.flava}}$	-0.01 [-0.07, 0.04]
$\beta^{\text{previous}}_{\text{species}_{B.alleghaniensis}}$	0.15 [0.02, 0.28]
$\beta^{\text{previous}}_{\text{species}_{B.nigra}}$	-0.08 [-0.17, -0.00]
$\beta^{\text{previous}}_{\text{species}_{C.glabra}}$	0.01 [-0.13, 0.15]
$\beta^{\text{previous}}_{\text{species}_{C.ovata}}$	-0.04 [-0.17, 0.09]
$\beta^{\text{previous}}_{\text{species}_{P.deltoides}}$	0.02 [-0.05, 0.10]
$\beta^{\text{previous}}_{\text{species}_{Q.alba}}$	-0.03 [-0.38, 0.33]
$\beta^{\text{previous}}_{\text{species}_{Q.rubra}}$	0.09 [-0.27, 0.46]
$\beta^{\text{previous}}_{\text{species}_{T.americana}}$	0.00 [-0.06, 0.06]
$\alpha_{\text{species}_{A.rubrum}}$	0.44 [-3.10, 3.90]
$\alpha_{\text{species}_{A.saccharum}}$	-2.20 [-4.79, 0.36]
$\alpha_{\text{species}_{Ae.flava}}$	0.04 [-1.53, 1.59]
$\alpha_{\text{species}_{B.alleghaniensis}}$	-0.06 [-2.61, 2.44]
$\alpha_{\text{species}_{B.nigra}}$	0.30 [-1.39, 1.99]
$\alpha_{\text{species}_{C.glabra}}$	0.18 [-1.77, 2.13]
$\alpha_{\text{species}_{C.ovata}}$	1.16 [-1.18, 3.47]
$\alpha_{\text{species}_{P.deltoides}}$	-0.01 [-1.92, 1.90]
$\alpha_{\text{species}_{Q.alba}}$	0.97 [-5.55, 7.68]
$\alpha_{\text{species}_{Q.rubra}}$	-1.73 [-8.12, 4.60]
$\alpha_{\text{species}_{T.americana}}$	0.93 [-0.65, 2.50]

Table A.10: Summary output of each parameter from the model on the effect of SOS (days) on EOS (days; Equation 2.3). We report estimates as mean and 90% uncertainty intervals (in square brackets). see 'Data analyses' section in Methods for more details

Parameter	Estimate
α	29.51 [23.54, 35.39]
$\sigma_{\alpha_{treeid}}$	1.18 [0.83, 1.56]
σ_y	2.06 [1.90, 2.23]
$\beta_{\text{species}_{A.rubrum}}$	0.63 [-0.14, 1.42]
$\beta_{\text{species}_{A.saccharum}}$	0.18 [-0.29, 0.66]
$\beta_{\text{species}_{Ae.flava}}$	0.38 [-0.32, 1.08]
$\beta_{\text{species}_{B.alleghaniensis}}$	0.50 [-0.15, 1.16]
$\beta_{\text{species}_{B.nigra}}$	0.44 [-0.16, 1.06]
$\beta_{\text{species}_{C.glabra}}$	0.48 [-0.17, 1.15]
$\beta_{\text{species}_{C.ovata}}$	0.40 [-0.22, 1.01]
$\beta_{\text{species}_{P.deltoides}}$	1.04 [0.39, 1.70]
$\beta_{\text{species}_{Q.alba}}$	0.35 [-0.34, 1.04]
$\beta_{\text{species}_{Q.rubra}}$	0.50 [-0.19, 1.22]
$\beta_{\text{species}_{T.americana}}$	0.46 [-0.14, 1.05]
$\alpha_{\text{species}_{A.rubrum}}$	-2.26 [-15.00, 10.18]
$\alpha_{\text{species}_{A.saccharum}}$	7.21 [-1.83, 16.23]
$\alpha_{\text{species}_{Ae.flava}}$	0.36 [-10.75, 11.25]
$\alpha_{\text{species}_{B.alleghaniensis}}$	-0.23 [-11.38, 10.91]
$\alpha_{\text{species}_{B.nigra}}$	0.27 [-10.43, 10.84]
$\alpha_{\text{species}_{C.glabra}}$	-0.20 [-11.26, 10.69]
$\alpha_{\text{species}_{C.ovata}}$	1.92 [-9.19, 13.03]
$\alpha_{\text{species}_{P.deltoides}}$	-11.53 [-23.17, 0.25]
$\alpha_{\text{species}_{Q.alba}}$	3.61 [-8.48, 15.48]
$\alpha_{\text{species}_{Q.rubra}}$	0.14 [-12.16, 12.19]
$\alpha_{\text{species}_{T.americana}}$	-0.27 [-10.42, 9.76]

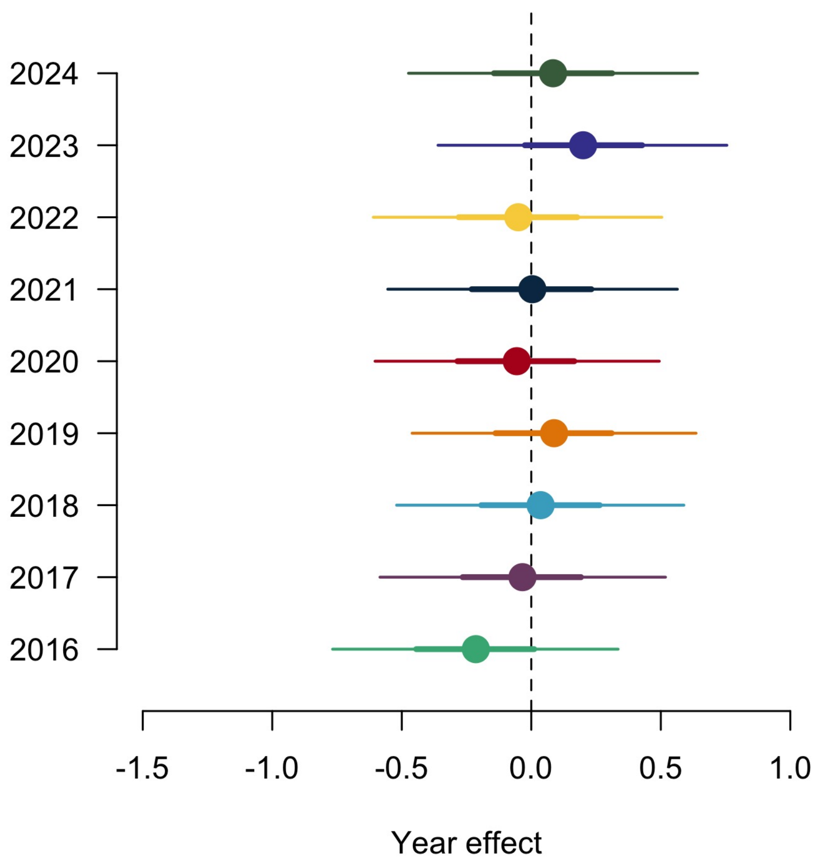

Figure A.13: Ring width (log(mm)) deviations from the grand mean for each year using the GDD model (Table A.8). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

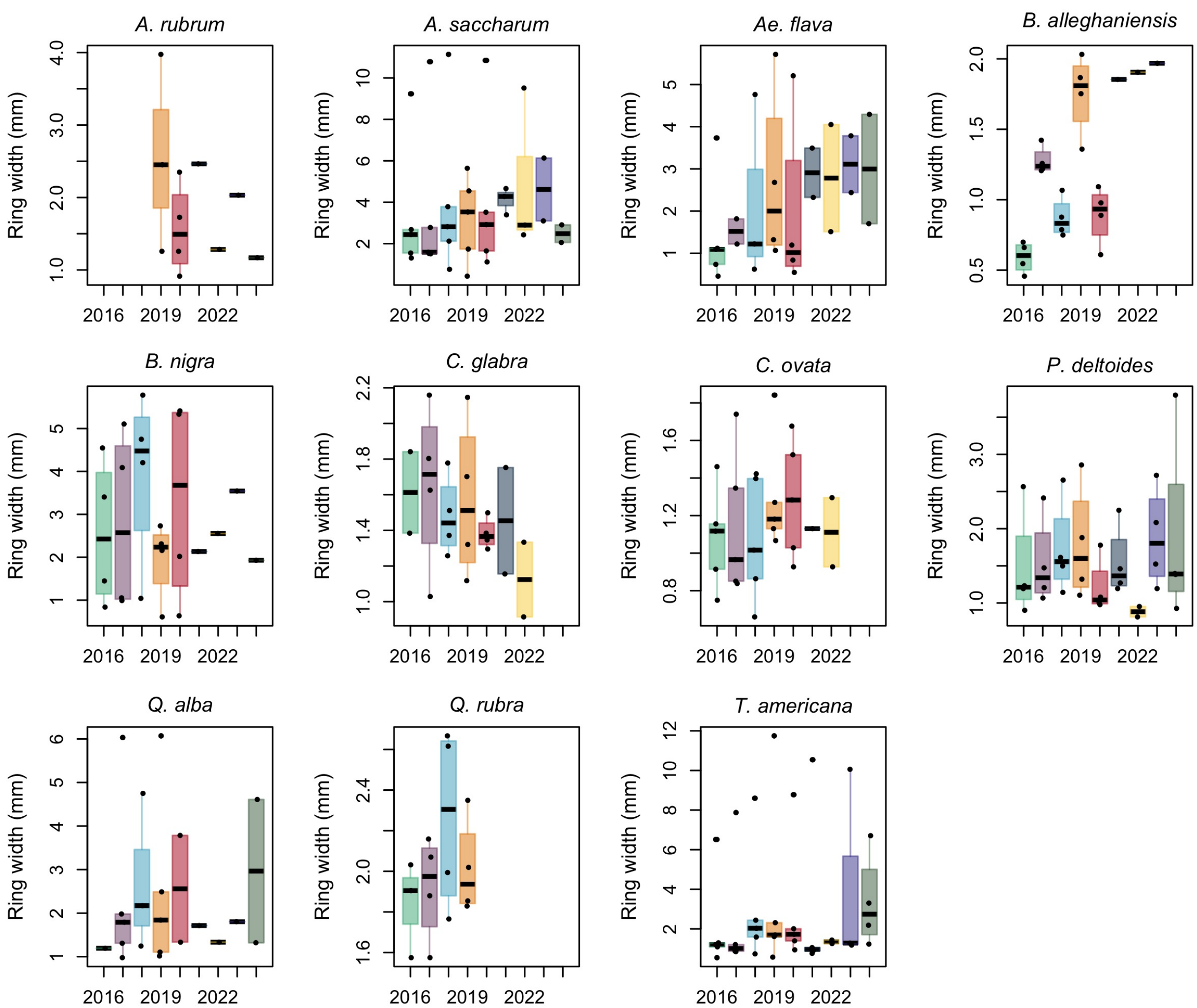

Figure A.14: Boxplot representing the empirical data of our ring widths (mm) across years, facetted by species. The dots represent the raw data, the line the mean and the box the interquantile range (IQR).

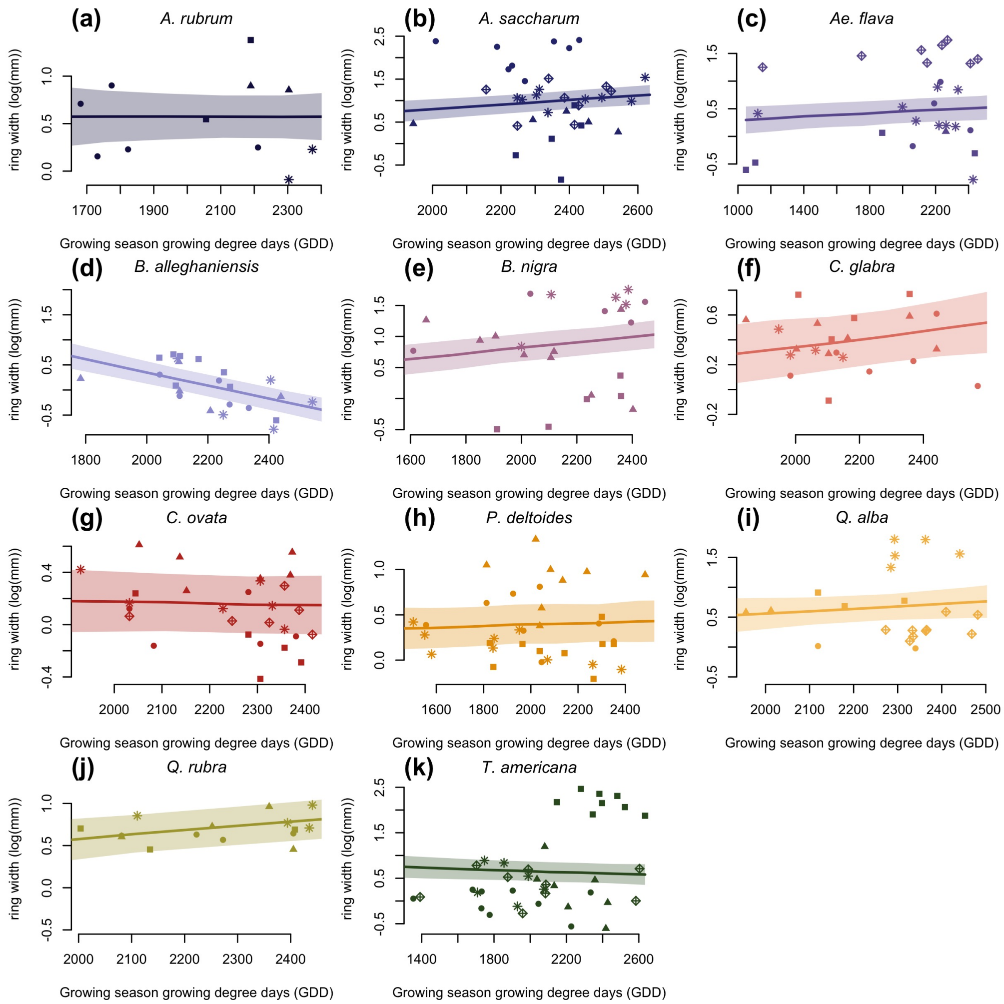

Figure A.15: Ring width (log(mm)) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

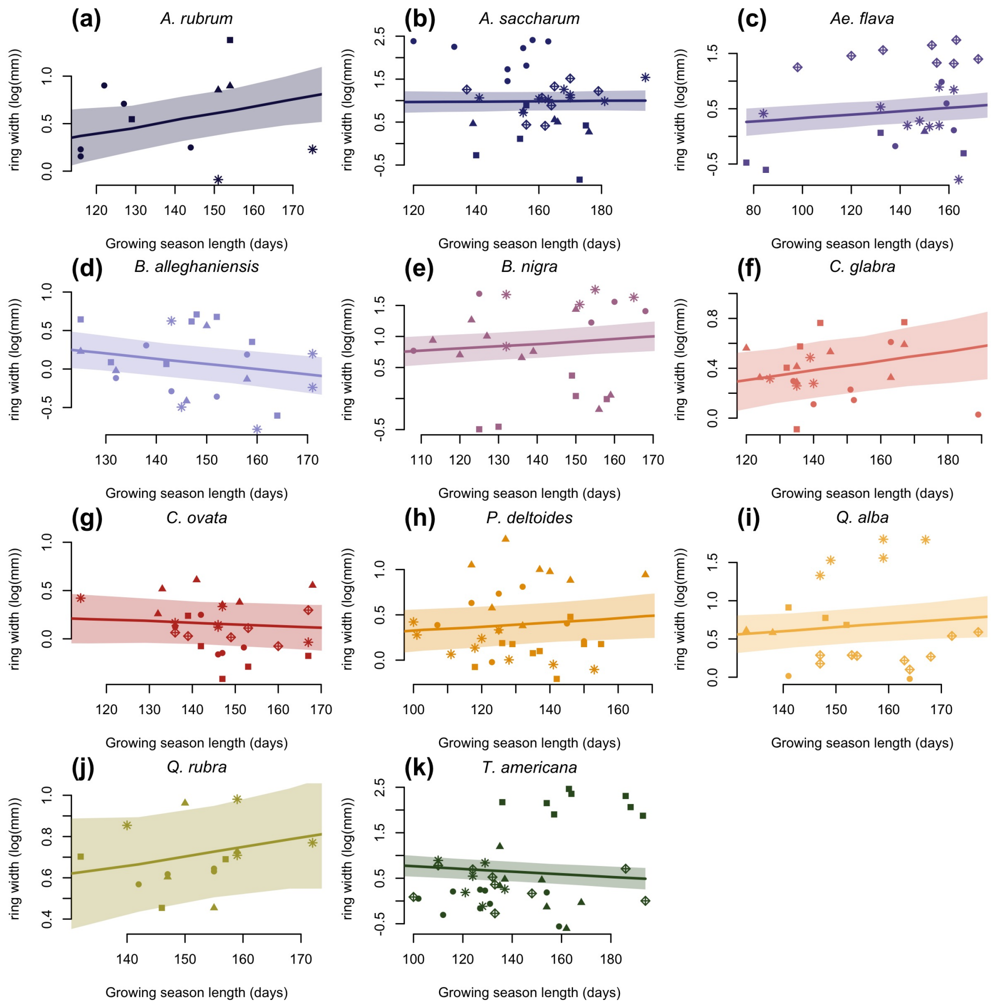

Figure A.16: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

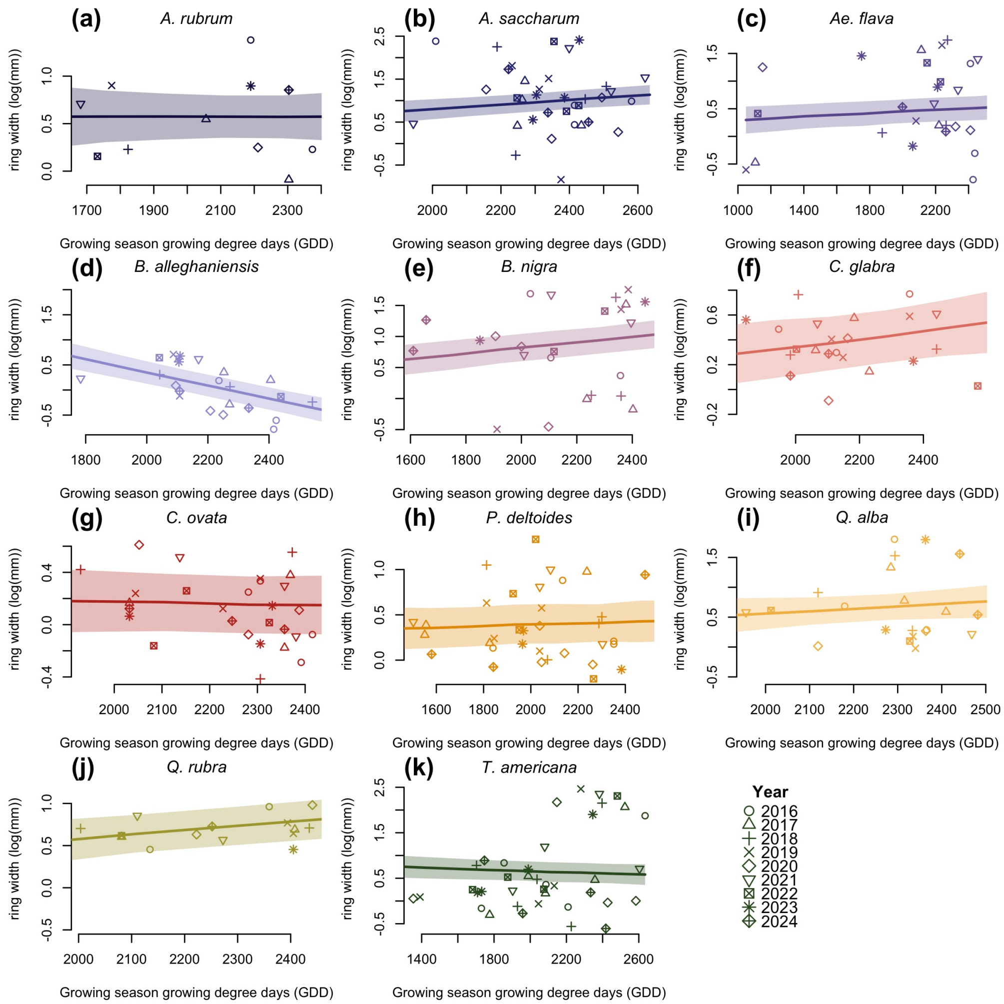

Figure A.17: Ring width trends (log(mm)) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

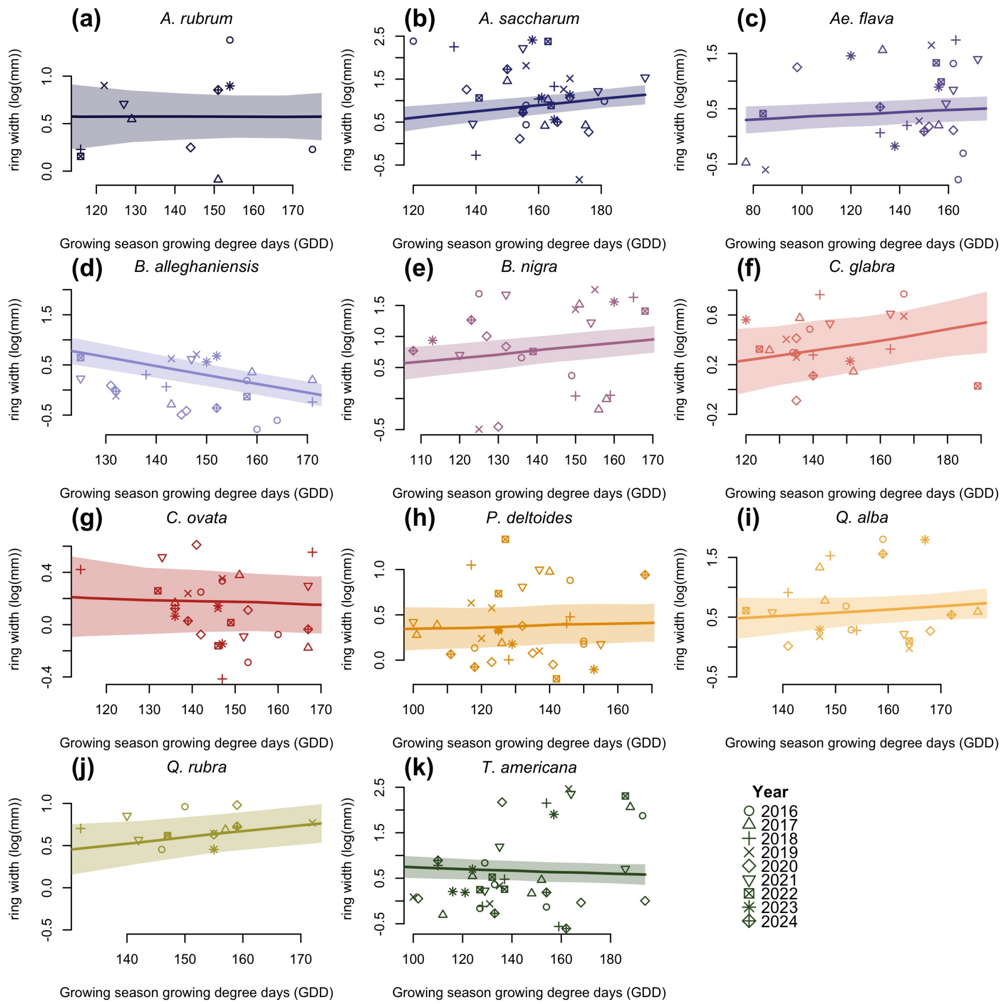

Figure A.18: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

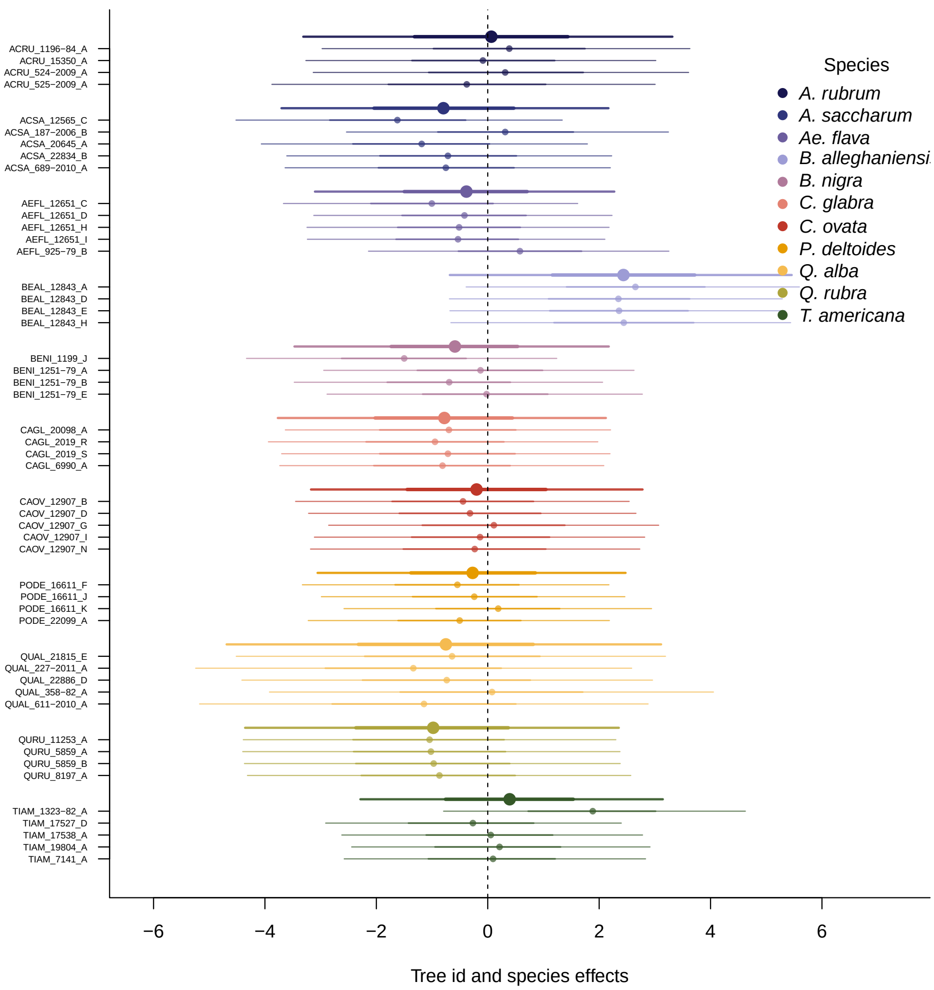

Figure A.19: Ring width (log(mm)) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts, which are the species-level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

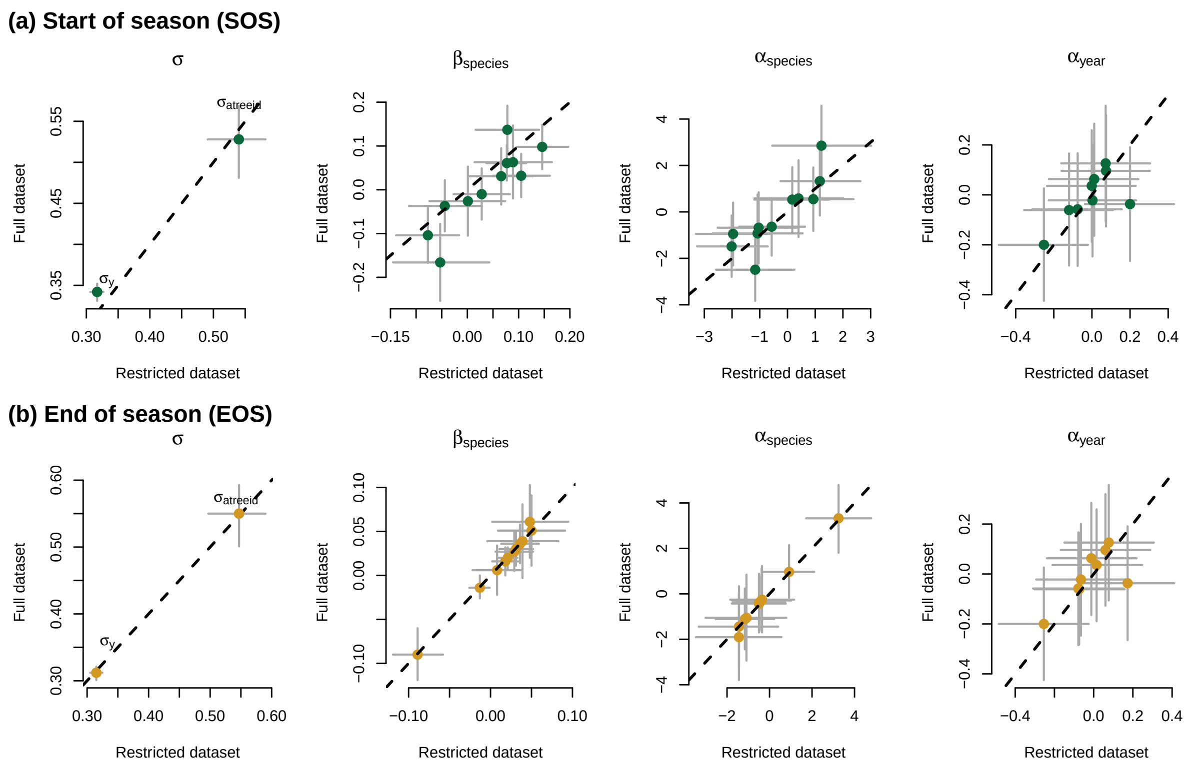

Figure A.20: Full vs, most restricted rows of data for the model with start of season (SOS; a) and end of season (EOS; b) as the predictor. Each panel represent the different parameters used to fit the models. $\alpha_{species}$ and α_{year} represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ represent the change in ring width (log(mm)) per scaled predictor for each species (see 'Data analyses' section in Methods for more details on predictor scaling).

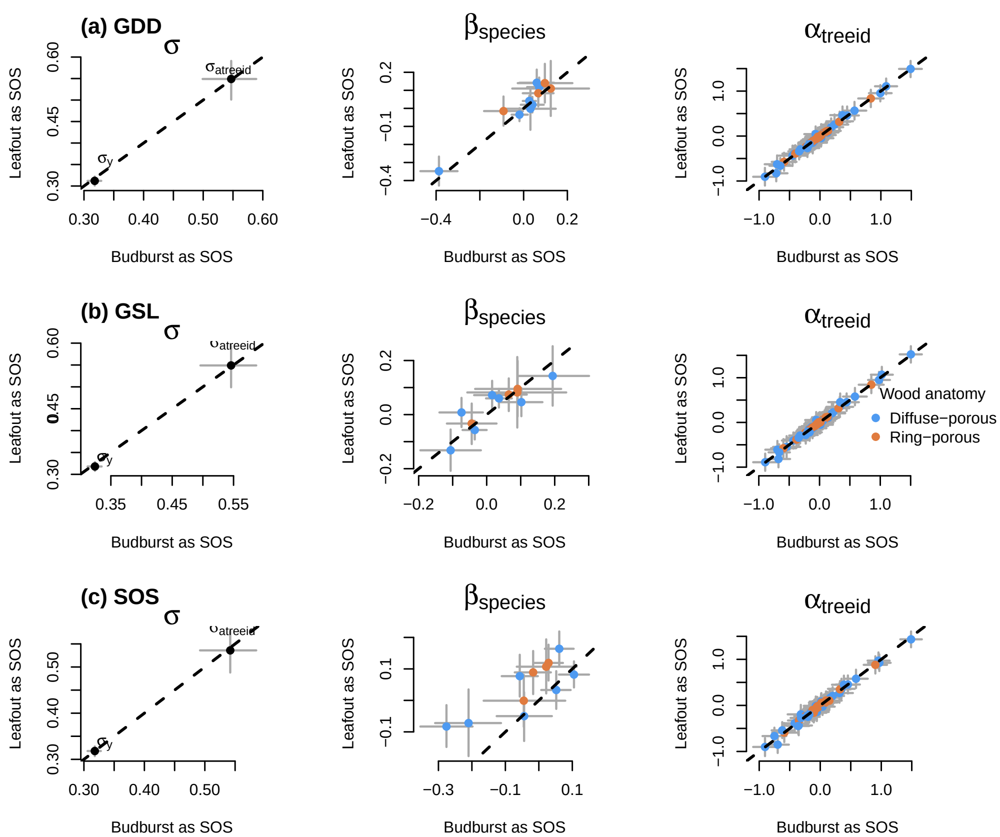

Figure A.21: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see 'Data analyses' section in Methods for more details).

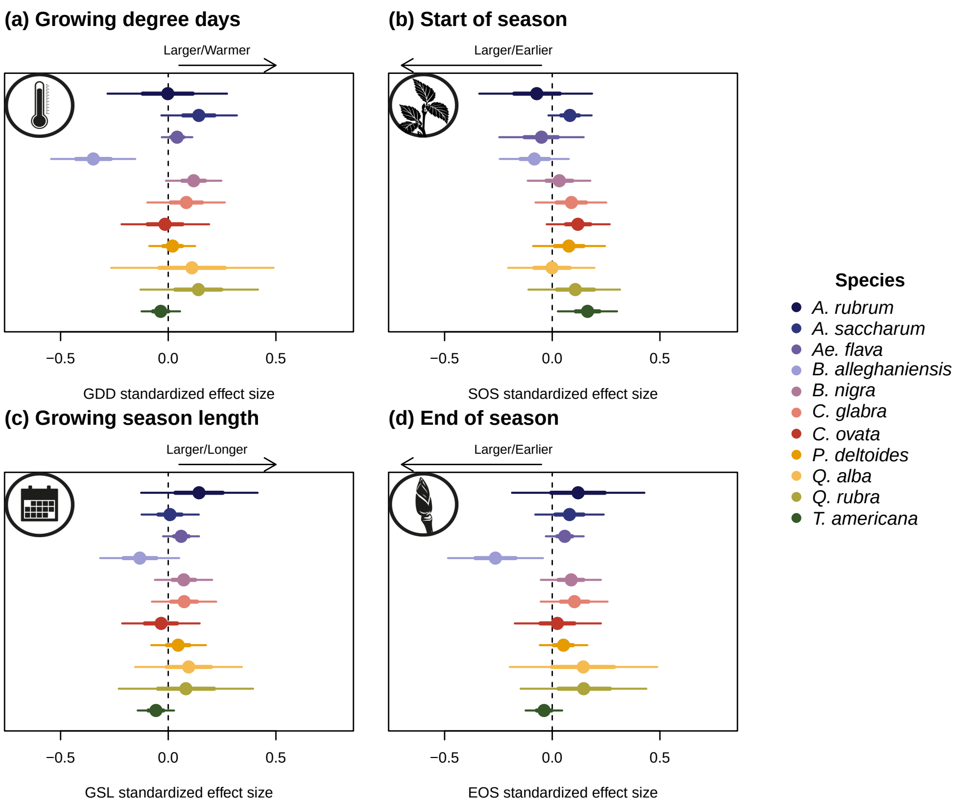

Figure A.22: Effect size of each predictor across our eleven species. We depict ring width (log(mm)) change per scaled (z-scored; see 'Data analyses' section in Methods for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

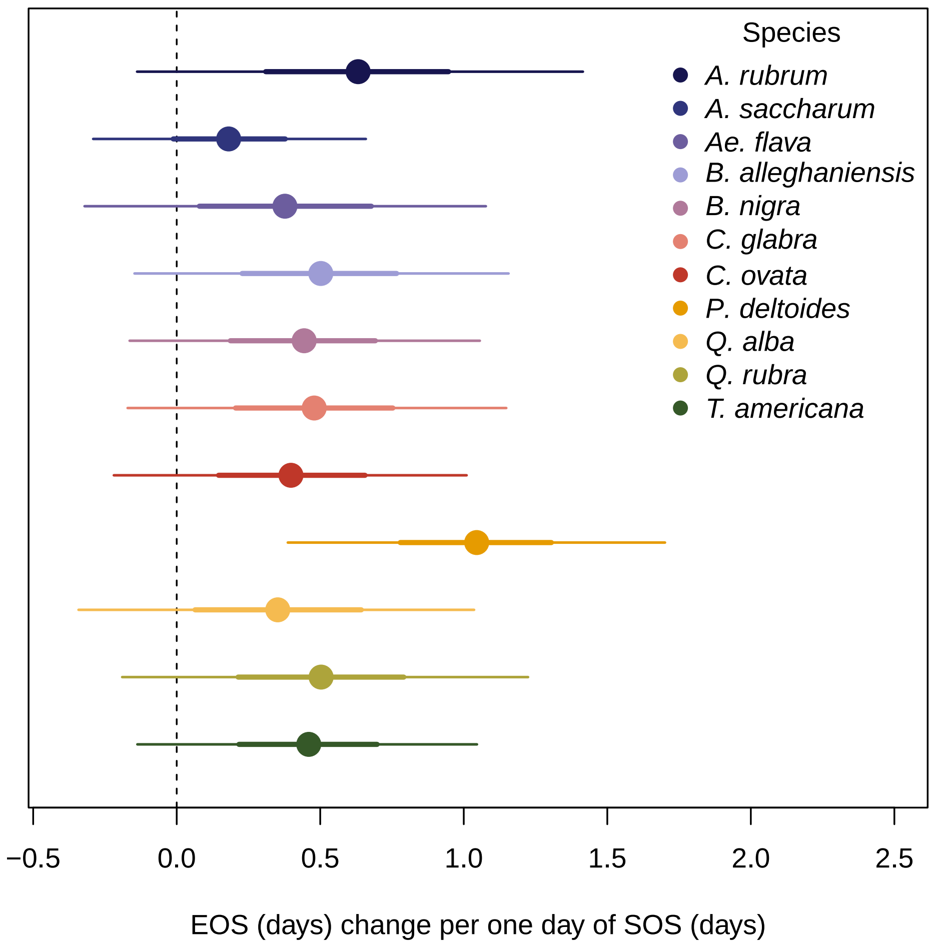

Figure A.23: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.